

Snippet 1: Chronological Time Split (Step 1)

This snippet handles the strict timeline splitting required for time-series data to guarantee absolutely zero future data leaks into the training sets.

```
import pandas as pd

# 1. Load telemetry data containing native columns: time, counts,
hardness
df = pd.read_csv("/content/combined_solexs_dataset.csv")

# 2. Enforce explicit temporal sorting to preserve timeline sequence
order
df['time'] = pd.to_datetime(df['time'])
df = df.sort_values('time').reset_index(drop=True)

# 3. Define sequential partitioning thresholds (70% Train, 15% Val,
15% Test)
total_rows = len(df)
train_end_idx = int(total_rows * 0.70)
val_end_idx = int(total_rows * 0.85)

print(f"Data split coordinates calculated successfully:")
print(f"Train rows: 0 to {train_end_idx} | Val rows: {train_end_idx}
to {val_end_idx} | Test rows: {val_end_idx} to {total_rows}")

Data split coordinates calculated successfully:
Train rows: 0 to 6652800 | Val rows: 6652800 to 8078400 | Test rows:
8078400 to 9504000
```

Snippet 2: Dynamic is_flare & Proactive Target Y Generation (Step 2)

This implements the 3-stage feature lifecycle (Stages 1 & 2), dynamically computing the moving statistical nowcasting threshold from the raw input and sliding it upside down to establish the 15-minute look-ahead horizon target array.

```
# Stage 1 (Instantiation): deriving the threshold flag dynamically via
code
flux_mean = df['counts'].mean()
flux_std = df['counts'].std()
nowcast_threshold = flux_mean + (5 * flux_std)
df['is_flare'] = (df['counts'] > nowcast_threshold).astype(int)

# Target Horizon Forecast Window: N = 15 minutes (15 mins * 60 seconds
= 900 rows)
look_ahead_steps = 15 * 60

# Stage 2 (Target Computation): Reverse rolling max to tag the
precursor zones
df['Y'] = (
```

```

df['is_flare']
  .iloc[::-1]
  .rolling(window=look_ahead_steps, min_periods=1)
  .max()
  .iloc[::-1]
  .fillna(0)
  .astype(int)
)

print(f"Dynamic target matching completed. Target class distribution:\n{df['Y'].value_counts()}")

Dynamic target matching completed. Target class distribution:
Y
0    9459825
1      44175
Name: count, dtype: int64

```

Snippet 3: Continuous Look-Backward Window Feature Matrix Extraction (Step 3)

This snippet slides a look-backward window ($W=900$ rows) along the continuous timeline to extract rolling statistical anomalies, multi-scale acceleration curves, and sub-band spectral fluctuations from both counts and hardness. It also implements Stage 3 of the lifecycle by strictly purging the nowcast variables to prevent training leaks.

```

# Look-backward summary context window W = 15 minutes (900 seconds)
W_steps = 15 * 60

features_dict = {
    # --- Thermal Soft X-Ray Volume Profiles ---
    'soft_energy_integral': df['counts'].rolling(window=W_steps,
min_periods=1).sum(),
    'soft_flux_variance': df['counts'].rolling(window=W_steps,
min_periods=1).var(),
    'soft_flux_std': df['counts'].rolling(window=W_steps,
min_periods=1).std(),
    'thermal_range_envelope': df['counts'].rolling(window=W_steps,
min_periods=1).max() - df['counts'].rolling(window=W_steps,
min_periods=1).min(),

    # --- Thermal Acceleration Momentum Derivatives ---
    'instantaneous_velocity_1s': df['counts'].diff(),
    'sustained_acceleration_1m': df['counts'].diff(periods=60)
}

# Integrate the sub-band spectral hardness ratio dynamics from the CSV
if present
if 'hardness' in df.columns:
    features_dict['hardness_rolling_mean'] =
df['hardness'].rolling(window=W_steps, min_periods=1).mean()

```

```

        features_dict['hardness_rolling_max'] =
df['hardness'].rolling(window=W_steps, min_periods=1).max()
        features_dict['hardness_moment_derivative'] =
df['hardness'].diff()

# Construct the initial matrix
X_matrix = pd.DataFrame(features_dict).fillna(0)
Y_vector = df['Y']

# Stage 3 (Deletion): CRITICAL LEAKAGE SHIELD
# splits are extracted without letting is_flare or Y bleed into the
# feature input space
X_train, Y_train = X_matrix.iloc[:train_end_idx],
Y_vector.iloc[:train_end_idx]
X_val, Y_val = X_matrix.iloc[train_end_idx:val_end_idx],
Y_vector.iloc[train_end_idx:val_end_idx]
X_test, Y_test = X_matrix.iloc[val_end_idx:],
Y_vector.iloc[val_end_idx:]

```

Snippet 4: Standard Scaling & Orthogonal Dimensional Compression (Step 4)

This snippet normalizes the feature variances and executes Principal Component Analysis (PCA) to collapse collinear, overlapping moving window trends into clean, uncorrelated component axes explaining 95% of the information space.

```

from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

# Apply standard scaling mapped strictly from the training metrics
# block
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)

# Validation Protection: ONLY transform validation and test splits.
# Never call .fit()!
X_val_scaled = scaler.transform(X_val)
X_test_scaled = scaler.transform(X_test)

# Map features onto independent axes retaining 95% total variance
# information
pca = PCA(n_components=0.95)
X_train_pca = pca.fit_transform(X_train_scaled)
X_val_pca = pca.transform(X_val_scaled)
X_test_pca = pca.transform(X_test_scaled)

print(f"PCA complete. Matrix collapsed from {X_train.shape[1]} raw
columns to {X_train_pca.shape[1]} principal components.")

PCA complete. Matrix collapsed from 9 raw columns to 6 principal
components.

```

```

from xgboost import XGBClassifier
from sklearn.metrics import classification_report, roc_auc_score

# 1. Calculate inverse class distribution ratio to counteract extreme
imbalance (<2%)
neg_count = (Y_train == 0).sum()
pos_count = (Y_train == 1).sum()
imbalance_ratio = neg_count / max(1, pos_count)

# 2. Instantiate the classifier optimized for space weather time-
series anomalies
xgb_model = XGBClassifier(
    n_estimators=120,
    max_depth=5,
    learning_rate=0.04,
    scale_pos_weight=imbalance_ratio, # Severe minority class
correction penalty
    eval_metric='logloss',
    random_state=42
)

# 3. Train the tree network securely monitored by the validation loss
tracking loops
xgb_model.fit(
    X_train_pca, Y_train,
    eval_set=[(X_val_pca, Y_val)],
    verbose=False
)

# 4. Generate prediction probabilities and binary classes for BOTH
splits
train_preds = xgb_model.predict(X_train_pca)
train_probs = xgb_model.predict_proba(X_train_pca)[:, 1]

val_preds = xgb_model.predict(X_val_pca)
val_probs = xgb_model.predict_proba(X_val_pca)[:, 1]

# 5. Output the Dual-Partition Evaluation Matrix
print("=====
=====")
print("
                                TRAINING DATASET PERFORMANCE MATRIX
")
print("=====
=====")
print(classification_report(Y_train, train_preds, target_names=["Quiet
(0)", "Precursor (1)"]))
print(f"Train ROC-AUC Score: {roc_auc_score(Y_train,
train_probs):.4f}\n")

print("=====
=====")

```

```

=====
print("
")
print("=====
=====")
print(classification_report(Y_val, val_preds, target_names=["Quiet
(0)", "Precursor (1)"]))
print(f"Validation ROC-AUC Score: {roc_auc_score(Y_val,
val_probs):.4f}")
print("=====
=====")

```

TRAINING DATASET PERFORMANCE MATRIX				
	precision	recall	f1-score	support
Quiet (0)	1.00	0.95	0.98	6618805
Precursor (1)	0.09	0.93	0.17	33995
accuracy			0.95	6652800
macro avg	0.55	0.94	0.57	6652800
weighted avg	1.00	0.95	0.97	6652800

Train ROC-AUC Score: 0.9882

VALIDATION DATASET PERFORMANCE MATRIX				
	precision	recall	f1-score	support
Quiet (0)	1.00	0.96	0.98	1418740
Precursor (1)	0.07	0.54	0.12	6860
accuracy			0.96	1425600
macro avg	0.53	0.75	0.55	1425600
weighted avg	0.99	0.96	0.98	1425600

Validation ROC-AUC Score: 0.8623

```

import lightgbm as lgb
from sklearn.metrics import classification_report, roc_auc_score

# 1. Calculate inverse class distribution ratio for imbalance
correction
neg_count = (Y_train == 0).sum()
pos_count = (Y_train == 1).sum()
imbalance_ratio = neg_count / max(1, pos_count)

# 2. Instantiate using lgb.LGBMClassifier to completely bypass the

```

```

wrapper bug
lgb_model = lgb.LGBMClassifier(
    n_estimators=120,
    max_depth=5,
    learning_rate=0.04,
    scale_pos_weight=imbalance_ratio, # Direct penalty multiplier for
minority precursor rows
    random_state=42,
    verbose=-1
)

# 3. Train the model
lgb_model.fit(X_train_pca, Y_train)

# 4. Generate predictions and floating probabilities for BOTH splits
train_preds = lgb_model.predict(X_train_pca)
train_probs = lgb_model.predict_proba(X_train_pca)[:, 1]

val_preds = lgb_model.predict(X_val_pca)
val_probs = lgb_model.predict_proba(X_val_pca)[:, 1]

# 5. Output the Dual-Partition Evaluation Matrix
print("=====
=====")
print("                                LIGHTGBM: TRAINING PARTITION METRICS
")
print("=====
=====")
print(classification_report(Y_train, train_preds, target_names=["Quiet
(0)", "Precursor (1)"]))
print(f"Train ROC-AUC Score: {roc_auc_score(Y_train,
train_probs):.4f}\n")

print("=====
=====")
print("                                LIGHTGBM: VALIDATION PARTITION METRICS
")
print("=====
=====")
print(classification_report(Y_val, val_preds, target_names=["Quiet
(0)", "Precursor (1)"]))
print(f"Validation ROC-AUC Score: {roc_auc_score(Y_val,
val_probs):.4f}")
print("=====
=====")

/usr/local/lib/python3.12/dist-packages/sklearn/utils/
validation.py:2739: UserWarning: X does not have valid feature names,
but LGBMClassifier was fitted with feature names
    warnings.warn(

```

```

/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:27
39: UserWarning: X does not have valid feature names, but
LGBMClassifier was fitted with feature names
warnings.warn(
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:27
39: UserWarning: X does not have valid feature names, but
LGBMClassifier was fitted with feature names
warnings.warn(
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:27
39: UserWarning: X does not have valid feature names, but
LGBMClassifier was fitted with feature names
warnings.warn(

```

```

=====
LIGHTGBM: TRAINING PARTITION METRICS
=====

```

	precision	recall	f1-score	support
Quiet (0)	1.00	0.95	0.97	6618805
Precursor (1)	0.08	0.96	0.15	33995
accuracy			0.95	6652800
macro avg	0.54	0.95	0.56	6652800
weighted avg	1.00	0.95	0.97	6652800

Train ROC-AUC Score: 0.9901

```

=====
LIGHTGBM: VALIDATION PARTITION METRICS
=====

```

	precision	recall	f1-score	support
Quiet (0)	1.00	0.96	0.98	1418740
Precursor (1)	0.07	0.55	0.12	6860
accuracy			0.96	1425600
macro avg	0.53	0.75	0.55	1425600
weighted avg	0.99	0.96	0.98	1425600

Validation ROC-AUC Score: 0.8256

```

#
=====
# SYSTEM DEPENDENCY INITIALIZATION
#
=====
print("Installing CatBoost framework package...")
# Install catboost quietly to keep your Colab workspace logs clean

```

```
!pip install -q catboost
```

```
print(" CatBoost successfully installed and ready for ensemble  
integration!")
```

```
Installing CatBoost framework package...
```

```
 CatBoost successfully installed and ready for ensemble integration!
```

```
from catboost import CatBoostClassifier  
from sklearn.metrics import classification_report, roc_auc_score
```

```
# 1. Calculate inverse class distribution ratio for scaling weights
```

```
neg_count = (Y_train == 0).sum()
```

```
pos_count = (Y_train == 1).sum()
```

```
imbalance_ratio = neg_count / max(1, pos_count)
```

```
# 2. Instantiate CatBoost utilizing scale_pos_weight for target  
alignment
```

```
cat_model = CatBoostClassifier(  
    iterations=120,  
    depth=5,  
    learning_rate=0.04,  
    scale_pos_weight=imbalance_ratio, # Penalizes missed precursor  
flags  
    random_seed=42,  
    verbose=0  
)
```

```
# 3. Train the model
```

```
cat_model.fit(X_train_pca, Y_train)
```

```
# 4. Generate predictions and floating probabilities for BOTH splits
```

```
train_preds = cat_model.predict(X_train_pca)
```

```
train_probs = cat_model.predict_proba(X_train_pca)[: , 1]
```

```
val_preds = cat_model.predict(X_val_pca)
```

```
val_probs = cat_model.predict_proba(X_val_pca)[: , 1]
```

```
# 5. Output the Dual-Partition Evaluation Matrix
```

```
print("=====  
=====")
```

```
print(" CATBOOST: TRAINING PARTITION METRICS  
")
```

```
print("=====  
=====")
```

```
print(classification_report(Y_train, train_preds, target_names=["Quiet  
(0)", "Precursor (1)"]))
```

```
print(f"Train ROC-AUC Score: {roc_auc_score(Y_train,  
train_probs):.4f}\n")
```



```

print("=====
=====")
print("
          CATBOOST: VALIDATION PARTITION METRICS
")
print("=====
=====")
print(classification_report(Y_val, val_preds, target_names=["Quiet
(0)", "Precursor (1)"]))
print(f"Validation ROC-AUC Score: {roc_auc_score(Y_val,
val_probs):.4f}")
print("=====
=====")

```

CATBOOST: TRAINING PARTITION METRICS				
	precision	recall	f1-score	support
Quiet (0)	1.00	0.94	0.97	6618805
Precursor (1)	0.07	0.91	0.14	33995
accuracy			0.94	6652800
macro avg	0.54	0.93	0.55	6652800
weighted avg	0.99	0.94	0.97	6652800

Train ROC-AUC Score: 0.9820

CATBOOST: VALIDATION PARTITION METRICS				
	precision	recall	f1-score	support
Quiet (0)	1.00	0.96	0.98	1418740
Precursor (1)	0.06	0.53	0.10	6860
accuracy			0.96	1425600
macro avg	0.53	0.74	0.54	1425600
weighted avg	0.99	0.96	0.97	1425600

Validation ROC-AUC Score: 0.8568

```

from sklearn.metrics import classification_report, roc_auc_score

#
=====
# PART 1: TRAINING DATASET METRICS (Zero Retraining Lag)
#
=====
print("Extracting base floating probabilities on Training Data...")

```

```

xgb_train_probs = xgb_model.predict_proba(X_train_pca)[: , 1]
lgb_train_probs = lgb_model.predict_proba(X_train_pca)[: , 1]
cat_train_probs = cat_model.predict_proba(X_train_pca)[: , 1]

# Mathematically pool and average the training probabilities
ensemble_train_probs = (xgb_train_probs + lgb_train_probs +
cat_train_probs) / 3.0
ensemble_train_preds = (ensemble_train_probs >= 0.50).astype(int)

# Print Training performance metrics
print("=====
=====")
print("                MANUAL ENSEMBLE: TRAINING PARTITION METRICS
")
print("=====
=====")
print(classification_report(Y_train, ensemble_train_preds,
target_names=["Quiet (0)", "Precursor (1)"]))
print(f"Ensemble Train ROC-AUC Score: {roc_auc_score(Y_train,
ensemble_train_probs):.4f}\n")

#
=====
# PART 2: VALIDATION DATASET METRICS (Zero Retraining Lag)
#
=====
print("Extracting base floating probabilities on Validation Data...")
xgb_val_probs = xgb_model.predict_proba(X_val_pca)[: , 1]
lgb_val_probs = lgb_model.predict_proba(X_val_pca)[: , 1]
cat_val_probs = cat_model.predict_proba(X_val_pca)[: , 1]

# Mathematically pool and average the validation probabilities
ensemble_val_probs = (xgb_val_probs + lgb_val_probs + cat_val_probs) /
3.0
ensemble_val_preds = (ensemble_val_probs >= 0.50).astype(int)

# Print Validation performance metrics
print("=====
=====")
print("                MANUAL ENSEMBLE: VALIDATION PARTITION METRICS
")
print("=====
=====")
print(classification_report(Y_val, ensemble_val_preds,
target_names=["Quiet (0)", "Precursor (1)"]))
print(f"Ensemble Validation ROC-AUC Score: {roc_auc_score(Y_val,
ensemble_val_probs):.4f}")
print("=====
=====")

```

Extracting base floating probabilities on Training Data...

```
/usr/local/lib/python3.12/dist-packages/sklearn/utils/
validation.py:2739: UserWarning: X does not have valid feature names,
but LGBMClassifier was fitted with feature names
  warnings.warn(
```

=====

MANUAL ENSEMBLE: TRAINING PARTITION METRICS

=====

	precision	recall	f1-score	support
Quiet (0)	1.00	0.95	0.98	6618805
Precursor (1)	0.09	0.94	0.17	33995
accuracy			0.95	6652800
macro avg	0.55	0.94	0.57	6652800
weighted avg	1.00	0.95	0.97	6652800

Ensemble Train ROC-AUC Score: 0.9890

Extracting base floating probabilities on Validation Data...

```
/usr/local/lib/python3.12/dist-packages/sklearn/utils/
validation.py:2739: UserWarning: X does not have valid feature names,
but LGBMClassifier was fitted with feature names
  warnings.warn(
```

=====

MANUAL ENSEMBLE: VALIDATION PARTITION METRICS

=====

	precision	recall	f1-score	support
Quiet (0)	1.00	0.96	0.98	1418740
Precursor (1)	0.06	0.54	0.11	6860
accuracy			0.96	1425600
macro avg	0.53	0.75	0.55	1425600
weighted avg	0.99	0.96	0.98	1425600

Ensemble Validation ROC-AUC Score: 0.8537

=====

```
import warnings
from sklearn.metrics import classification_report, roc_auc_score

print("Extracting base floating probabilities on Validation Data...")
# Wrap predictions in a warning filter to keep your Colab logs
perfectly clean for presentation screen captures
with warnings.catch_warnings():
    warnings.simplefilter("ignore", category=UserWarning)
```

```
xgb_val_probs = xgb_model.predict_proba(X_val_pca)[: , 1]
lgb_val_probs = lgb_model.predict_proba(X_val_pca)[: , 1]
cat_val_probs = cat_model.predict_proba(X_val_pca)[: , 1]
```

Extracting base floating probabilities on Validation Data...

PHASE 5

Updated Phase 5

```
import numpy as np
import pandas as pd
import warnings
import matplotlib.pyplot as plt
import seaborn as sns

# Configure presentation aesthetics matching aerospace engineering
reports
sns.set_theme(style="whitegrid")
plt.rcParams.update({'font.size': 11, 'axes.labelsize': 12,
'axes.titlesize': 14})

#
=====
# 1. SETUP CHRONOLOGICAL VALIDATION AND HIDDEN TEST DATA TIMELINES
#
=====
X_val = X_matrix.iloc[train_end_idx:val_end_idx]
val_times =
df['time'].iloc[train_end_idx:val_end_idx].reset_index(drop=True)
val_df = pd.DataFrame({
    'time': val_times,
    'counts':
df['counts'].iloc[train_end_idx:val_end_idx].reset_index(drop=True),
    'is_flare':
df['is_flare'].iloc[train_end_idx:val_end_idx].reset_index(drop=True)
})
val_df['flare_start'] = ((val_df['is_flare'] == 1) &
(val_df['is_flare'].shift(1) == 0)).astype(int)
val_flare_indices = val_df[val_df['flare_start'] == 1].index.tolist()

X_test = X_matrix.iloc[val_end_idx:]
test_times = df['time'].iloc[val_end_idx:].reset_index(drop=True)
test_df = pd.DataFrame({
    'time': test_times,
    'counts': df['counts'].iloc[val_end_idx:].reset_index(drop=True),
    'is_flare':
df['is_flare'].iloc[val_end_idx:].reset_index(drop=True)
```

```

}))
test_df['flare_start'] = ((test_df['is_flare'] == 1) &
(test_df['is_flare'].shift(1) == 0)).astype(int)
test_flare_indices = test_df[test_df['flare_start'] ==
1].index.tolist()

#
=====
# 2. EXTRACT PROBABILITIES FOR BOTH SECTIONS
#
=====
print("Extracting base ensemble soft probabilities across Validation
and Test sets...")
with warnings.catch_warnings():
    warnings.simplefilter("ignore", category=UserWarning)
    val_df['pred_prob'] = (xgb_model.predict_proba(X_val_pca)[: , 1] +
                           lgb_model.predict(X_val_pca) +
                           cat_model.predict_proba(X_val_pca)[: , 1]) /
3.0

    test_df['pred_prob'] = (xgb_model.predict_proba(X_test_pca)[: , 1]
+
                           lgb_model.predict(X_test_pca) +
                           cat_model.predict_proba(X_test_pca)[: , 1])
/ 3.0

#
=====
# 3. RUN THE 3D SEARCH SWEEP ON VALIDATION DATA
#
=====
candidate_thresholds = [0.97, 0.98, 0.99]
candidate_durations = list(range(5, 25)) # 6s up to 12s
candidate_windows = list(range(10, 121, 5)) # 30s, 40s, ..., 90s

HORIZON_SECONDS = 15 * 60
sweep_records = []

print("\n" + "-"*105)
print("    STAGE 1: RUNNING MULTIDIMENSIONAL 3D PARAMETRIC SWEEP
(THRESHOLD x DURATION x ROLLING WINDOW)")
print("-"*105)

# 3D Grid Loop
for window_length in candidate_windows:
    # Compute the raw binary crossing points for this threshold ahead
    of time to save iterations
    for threshold in candidate_thresholds:
        val_df['raw_cross'] = (val_df['pred_prob'] >=
threshold).astype(int)

```

```

        # Calculate the rolling cumulative sum dynamically for this
        window length step
        rolling_cross_sum =
val_df['raw_cross'].rolling(window=window_length, min_periods=1).sum()

        for active_secs in candidate_durations:
            # Skip physically impossible conditions where required
            crossing seconds exceed total window length
            if active_secs > window_length:
                continue

            val_df['alarm_triggered'] = (rolling_cross_sum >=
active_secs).astype(int)

            true_positives = 0
            false_negatives = 0
            matched_alarm_indices = set()

            for start_idx in val_flare_indices:
                window_start = max(0, start_idx - HORIZON_SECONDS)
                precursor_window = val_df.iloc[window_start:start_idx]
                triggers =
precursor_window[precursor_window['alarm_triggered'] == 1]

                if len(triggers) > 0:
                    true_positives += 1

            matched_alarm_indices.update(precursor_window[precursor_window['alarm_
triggered'] == 1].index.tolist())
            else:
                false_negatives += 1

            all_alarms = set(val_df[val_df['alarm_triggered'] ==
1].index.tolist())
            false_positive_rows =
all_alarms.difference(matched_alarm_indices)

            if len(false_positive_rows) > 0:
                fp_series =
pd.Series(list(false_positive_rows)).sort_values()
                false_positives_events = (fp_series.diff() > 60).sum()
+ 1
            else:
                false_positives_events = 0

            event_tpr = true_positives / max(1, (true_positives +
false_negatives))
            event_csi = true_positives / max(1, (true_positives +
false_positives_events + false_negatives))

```

```

        sweep_records.append({
            'window_size': window_length,
            'threshold': threshold,
            'duration_sec': active_secs,
            'caught': true_positives,
            'false_alarms': false_positives_events,
            'tpr': event_tpr,
            'csi': event_csi
        })

    print(f"Window: {window_length}s | Thresh: {threshold:.2f}
| Min Dur: {active_secs:2d}s -> "
        f"Caught: {true_positives:2d} | FA:
{false_positives_events:2d} | CSI: {event_csi:.4f}")

# Extract the optimal multi-variable configuration from validation
sweep_df = pd.DataFrame(sweep_records)
optimal_runs = sweep_df[sweep_df['tpr'] == sweep_df['tpr'].max()]
best_config = optimal_runs.loc[optimal_runs['csi'].idxmax()]

best_window = int(best_config['window_size'])
best_thresh = float(best_config['threshold'])
best_dur = int(best_config['duration_sec'])

print("-" * 105)
print(f"  3D SELECTION COMPLETE! Multi-Variable Validation Optimum
Identified:")
print(f" -> Chosen Rolling Window      : {best_window} Seconds")
print(f" -> Chosen Threshold Boundary : {best_thresh:.2f}")
print(f" -> Chosen Filter Duration    : {best_dur} Seconds")

#
=====
# 4. GRAPHICAL TREND LINE VISUALIZATION
#
=====
print("\nGenerating visual performance trend map...")
plt.figure(figsize=(10, 5))

# Group by rolling window sizes to observe overall system behavior on
CSI stability
window_trend = sweep_df.groupby('window_size')
['csi'].mean().reset_index()
plt.plot(window_trend['window_size'], window_trend['csi'],
color='#3498DB', marker='o', lw=3, label='Mean Validation CSI')

plt.axvline(x=best_window, color='crimson', linestyle=':', lw=2.5,
label=f"Selected Window Peak ({best_window}s)")
plt.title("Impact of Rolling Window Constraints on Pipeline CSI

```

```

Efficiency", weight='bold', pad=12)
plt.xlabel("Lookback Rolling Window Size (Seconds)")
plt.ylabel("Critical Success Index (CSI Score)")
plt.legend()
plt.tight_layout()
plt.show()

#
=====
# 5. DEPLOY THE WINNING COMBINATION ON THE HIDDEN TEST DATA
#
=====
print("\n" + "="*105)
print("  STAGE 2: DEPLOYING DYNAMIC 3D OPTIMIZED CONSTRAINTS TO BLIND
TEST SPLIT")
print("="*105)

test_df['raw_cross'] = (test_df['pred_prob'] >=
best_thresh).astype(int)
test_rolling_cross = test_df['raw_cross'].rolling(window=best_window,
min_periods=1).sum()
test_df['alarm_triggered'] = (test_rolling_cross >=
best_dur).astype(int)

test_tp = 0
test_fn = 0
lead_times_minutes = []
test_matched_alarm_indices = set()

for start_idx in test_flare_indices:
    end_search = min(start_idx + 1800, len(test_df))
    peak_idx = test_df.iloc[start_idx:end_search]['counts'].idxmax()

    window_start = max(0, start_idx - HORIZON_SECONDS)
    precursor_window = test_df.iloc[window_start:start_idx]
    triggers = precursor_window[precursor_window['alarm_triggered'] ==
1]

    if len(triggers) > 0:
        test_tp += 1
        first_trigger_idx = triggers.index[0]

test_matched_alarm_indices.update(precursor_window[precursor_window['a
alarm_triggered'] == 1].index.tolist())
    lead_times_minutes.append((peak_idx - first_trigger_idx) /
60.0)
    else:
        test_fn += 1

test_all_alarms = set(test_df[test_df['alarm_triggered'] ==

```



```

l].index.tolist())
test_false_positive_rows =
test_all_alarms.difference(test_matched_alarm_indices)

if len(test_false_positive_rows) > 0:
    test_fp_series =
pd.Series(list(test_false_positive_rows)).sort_values()
    test_fp_events = (test_fp_series.diff() > 60).sum() + 1
else:
    test_fp_events = 0

test_quiet_mask = (test_df['is_flare'] == 0) &
(~test_df.index.isin(test_matched_alarm_indices))
test_total_possible_fa_windows = max(1, int(test_quiet_mask.sum() /
HORIZON_SECONDS))

# Final Test Metrics Compilation
final_tpr = test_tp / max(1, (test_tp + test_fn))
final_far = test_fp_events / max(1, (test_tp + test_fp_events))
final_csi = test_tp / max(1, (test_tp + test_fp_events + test_fn))

print(f" Confirmed Flares Occurred      : {len(test_flare_indices)}")
print(f" Flares Successfully Anticipated : {test_tp} /
{len(test_flare_indices)}")
print(f" Flares Completely Missed         : {test_fn}")
print(f" Total Operational False Alarms   : {test_fp_events} (Out of
{test_total_possible_fa_windows}, Quiet Windows)")
print("-"*105)
print(f" □ UNBIASED FINAL TRUE POSITIVE RATE (TPR)   : {final_tpr *
100:.2f}%")
print(f" □ UNBIASED FINAL FALSE ALARM RATE (FAR)      : {final_far *
100:.2f}%")
print(f" □ UNBIASED FINAL CRITICAL SUCCESS INDEX(CSI):
{final_csi:.4f}")
print("-"*105)
if len(lead_times_minutes) > 0:
    print(f" Test Split Median Early Warning cushion      :
{np.median(lead_times_minutes):.2f} Minutes")
print("="*105)

```

Extracting base ensemble soft probabilities across Validation and Test sets...

```

-----
-----
STAGE 1: RUNNING MULTIDIMENSIONAL 3D PARAMETRIC SWEEP (THRESHOLD x
DURATION x ROLLING WINDOW)
-----
-----
Window: 10s | Thresh: 0.97 | Min Dur: 5s -> Caught: 22 | FA: 15 |

```

```

CSI: 0.5789
Window: 10s | Thresh: 0.97 | Min Dur: 6s -> Caught: 22 | FA: 12 |
CSI: 0.6286
Window: 10s | Thresh: 0.97 | Min Dur: 7s -> Caught: 21 | FA: 10 |
CSI: 0.6364
Window: 10s | Thresh: 0.97 | Min Dur: 8s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 10s | Thresh: 0.97 | Min Dur: 9s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 10s | Thresh: 0.97 | Min Dur: 10s -> Caught: 19 | FA: 7 |
CSI: 0.6333
Window: 10s | Thresh: 0.98 | Min Dur: 5s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 10s | Thresh: 0.98 | Min Dur: 6s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 10s | Thresh: 0.98 | Min Dur: 7s -> Caught: 19 | FA: 7 |
CSI: 0.6333
Window: 10s | Thresh: 0.98 | Min Dur: 8s -> Caught: 19 | FA: 7 |
CSI: 0.6333
Window: 10s | Thresh: 0.98 | Min Dur: 9s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 10s | Thresh: 0.98 | Min Dur: 10s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 10s | Thresh: 0.99 | Min Dur: 5s -> Caught: 15 | FA: 8 |
CSI: 0.4839
Window: 10s | Thresh: 0.99 | Min Dur: 6s -> Caught: 15 | FA: 8 |
CSI: 0.4839
Window: 10s | Thresh: 0.99 | Min Dur: 7s -> Caught: 15 | FA: 8 |
CSI: 0.4839
Window: 10s | Thresh: 0.99 | Min Dur: 8s -> Caught: 15 | FA: 8 |
CSI: 0.4839
Window: 10s | Thresh: 0.99 | Min Dur: 9s -> Caught: 15 | FA: 6 |
CSI: 0.5172
Window: 10s | Thresh: 0.99 | Min Dur: 10s -> Caught: 15 | FA: 6 |
CSI: 0.5172
Window: 15s | Thresh: 0.97 | Min Dur: 5s -> Caught: 22 | FA: 14 |
CSI: 0.5946
Window: 15s | Thresh: 0.97 | Min Dur: 6s -> Caught: 22 | FA: 13 |
CSI: 0.6111
Window: 15s | Thresh: 0.97 | Min Dur: 7s -> Caught: 22 | FA: 10 |
CSI: 0.6667
Window: 15s | Thresh: 0.97 | Min Dur: 8s -> Caught: 21 | FA: 12 |
CSI: 0.6000
Window: 15s | Thresh: 0.97 | Min Dur: 9s -> Caught: 19 | FA: 11 |
CSI: 0.5588
Window: 15s | Thresh: 0.97 | Min Dur: 10s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 15s | Thresh: 0.97 | Min Dur: 11s -> Caught: 19 | FA: 9 |
CSI: 0.5938

```

Window: 15s		Thresh: 0.97		Min Dur: 12s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 15s		Thresh: 0.97		Min Dur: 13s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 15s		Thresh: 0.97		Min Dur: 14s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 15s		Thresh: 0.97		Min Dur: 15s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 15s		Thresh: 0.98		Min Dur: 5s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 15s		Thresh: 0.98		Min Dur: 6s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 15s		Thresh: 0.98		Min Dur: 7s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 15s		Thresh: 0.98		Min Dur: 8s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 15s		Thresh: 0.98		Min Dur: 9s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 15s		Thresh: 0.98		Min Dur: 10s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 15s		Thresh: 0.98		Min Dur: 11s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 15s		Thresh: 0.98		Min Dur: 12s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 15s		Thresh: 0.98		Min Dur: 13s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 15s		Thresh: 0.98		Min Dur: 14s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 15s		Thresh: 0.98		Min Dur: 15s	->	Caught: 15		FA: 8	
CSI: 0.4839									
Window: 15s		Thresh: 0.99		Min Dur: 5s	->	Caught: 15		FA: 8	
CSI: 0.4839									
Window: 15s		Thresh: 0.99		Min Dur: 6s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 15s		Thresh: 0.99		Min Dur: 7s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 15s		Thresh: 0.99		Min Dur: 8s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 15s		Thresh: 0.99		Min Dur: 9s	->	Caught: 15		FA: 8	
CSI: 0.4839									
Window: 15s		Thresh: 0.99		Min Dur: 10s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 15s		Thresh: 0.99		Min Dur: 11s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 15s		Thresh: 0.99		Min Dur: 12s	->	Caught: 15		FA: 6	
CSI: 0.5172									
Window: 15s		Thresh: 0.99		Min Dur: 13s	->	Caught: 15		FA: 6	
CSI: 0.5172									
Window: 15s		Thresh: 0.99		Min Dur: 14s	->	Caught: 15		FA: 6	

CSI: 0.5172					
Window: 15s		Thresh: 0.99		Min Dur: 15s ->	Caught: 15 FA: 6
CSI: 0.5172					
Window: 20s		Thresh: 0.97		Min Dur: 5s ->	Caught: 22 FA: 13
CSI: 0.6111					
Window: 20s		Thresh: 0.97		Min Dur: 6s ->	Caught: 22 FA: 14
CSI: 0.5946					
Window: 20s		Thresh: 0.97		Min Dur: 7s ->	Caught: 22 FA: 12
CSI: 0.6286					
Window: 20s		Thresh: 0.97		Min Dur: 8s ->	Caught: 22 FA: 12
CSI: 0.6286					
Window: 20s		Thresh: 0.97		Min Dur: 9s ->	Caught: 21 FA: 10
CSI: 0.6364					
Window: 20s		Thresh: 0.97		Min Dur: 10s ->	Caught: 19 FA: 12
CSI: 0.5429					
Window: 20s		Thresh: 0.97		Min Dur: 11s ->	Caught: 19 FA: 12
CSI: 0.5429					
Window: 20s		Thresh: 0.97		Min Dur: 12s ->	Caught: 19 FA: 9
CSI: 0.5938					
Window: 20s		Thresh: 0.97		Min Dur: 13s ->	Caught: 19 FA: 9
CSI: 0.5938					
Window: 20s		Thresh: 0.97		Min Dur: 14s ->	Caught: 19 FA: 8
CSI: 0.6129					
Window: 20s		Thresh: 0.97		Min Dur: 15s ->	Caught: 19 FA: 8
CSI: 0.6129					
Window: 20s		Thresh: 0.97		Min Dur: 16s ->	Caught: 19 FA: 7
CSI: 0.6333					
Window: 20s		Thresh: 0.97		Min Dur: 17s ->	Caught: 19 FA: 7
CSI: 0.6333					
Window: 20s		Thresh: 0.97		Min Dur: 18s ->	Caught: 19 FA: 7
CSI: 0.6333					
Window: 20s		Thresh: 0.97		Min Dur: 19s ->	Caught: 19 FA: 7
CSI: 0.6333					
Window: 20s		Thresh: 0.97		Min Dur: 20s ->	Caught: 15 FA: 7
CSI: 0.5000					
Window: 20s		Thresh: 0.98		Min Dur: 5s ->	Caught: 19 FA: 9
CSI: 0.5938					
Window: 20s		Thresh: 0.98		Min Dur: 6s ->	Caught: 19 FA: 8
CSI: 0.6129					
Window: 20s		Thresh: 0.98		Min Dur: 7s ->	Caught: 19 FA: 8
CSI: 0.6129					
Window: 20s		Thresh: 0.98		Min Dur: 8s ->	Caught: 19 FA: 8
CSI: 0.6129					
Window: 20s		Thresh: 0.98		Min Dur: 9s ->	Caught: 19 FA: 8
CSI: 0.6129					
Window: 20s		Thresh: 0.98		Min Dur: 10s ->	Caught: 19 FA: 7
CSI: 0.6333					
Window: 20s		Thresh: 0.98		Min Dur: 11s ->	Caught: 19 FA: 7
CSI: 0.6333					

Window: 20s		Thresh: 0.98		Min Dur: 12s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 20s		Thresh: 0.98		Min Dur: 13s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 20s		Thresh: 0.98		Min Dur: 14s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 20s		Thresh: 0.98		Min Dur: 15s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 20s		Thresh: 0.98		Min Dur: 16s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 20s		Thresh: 0.98		Min Dur: 17s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 20s		Thresh: 0.98		Min Dur: 18s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 20s		Thresh: 0.98		Min Dur: 19s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 20s		Thresh: 0.98		Min Dur: 20s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 20s		Thresh: 0.99		Min Dur: 5s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 20s		Thresh: 0.99		Min Dur: 6s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 20s		Thresh: 0.99		Min Dur: 7s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 20s		Thresh: 0.99		Min Dur: 8s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 20s		Thresh: 0.99		Min Dur: 9s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 20s		Thresh: 0.99		Min Dur: 10s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 20s		Thresh: 0.99		Min Dur: 11s	->	Caught: 15		FA: 8	
CSI: 0.4839									
Window: 20s		Thresh: 0.99		Min Dur: 12s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 20s		Thresh: 0.99		Min Dur: 13s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 20s		Thresh: 0.99		Min Dur: 14s	->	Caught: 15		FA: 6	
CSI: 0.5172									
Window: 20s		Thresh: 0.99		Min Dur: 15s	->	Caught: 15		FA: 6	
CSI: 0.5172									
Window: 20s		Thresh: 0.99		Min Dur: 16s	->	Caught: 15		FA: 6	
CSI: 0.5172									
Window: 20s		Thresh: 0.99		Min Dur: 17s	->	Caught: 15		FA: 6	
CSI: 0.5172									
Window: 20s		Thresh: 0.99		Min Dur: 18s	->	Caught: 15		FA: 6	
CSI: 0.5172									
Window: 20s		Thresh: 0.99		Min Dur: 19s	->	Caught: 15		FA: 6	
CSI: 0.5172									
Window: 20s		Thresh: 0.99		Min Dur: 20s	->	Caught: 15		FA: 6	

```

CSI: 0.5172
Window: 25s | Thresh: 0.97 | Min Dur: 5s -> Caught: 22 | FA: 12 |
CSI: 0.6286
Window: 25s | Thresh: 0.97 | Min Dur: 6s -> Caught: 22 | FA: 12 |
CSI: 0.6286
Window: 25s | Thresh: 0.97 | Min Dur: 7s -> Caught: 22 | FA: 12 |
CSI: 0.6286
Window: 25s | Thresh: 0.97 | Min Dur: 8s -> Caught: 22 | FA: 12 |
CSI: 0.6286
Window: 25s | Thresh: 0.97 | Min Dur: 9s -> Caught: 21 | FA: 11 |
CSI: 0.6176
Window: 25s | Thresh: 0.97 | Min Dur: 10s -> Caught: 20 | FA: 11 |
CSI: 0.5882
Window: 25s | Thresh: 0.97 | Min Dur: 11s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 25s | Thresh: 0.97 | Min Dur: 12s -> Caught: 19 | FA: 11 |
CSI: 0.5588
Window: 25s | Thresh: 0.97 | Min Dur: 13s -> Caught: 19 | FA: 11 |
CSI: 0.5588
Window: 25s | Thresh: 0.97 | Min Dur: 14s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 25s | Thresh: 0.97 | Min Dur: 15s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 25s | Thresh: 0.97 | Min Dur: 16s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 25s | Thresh: 0.97 | Min Dur: 17s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 25s | Thresh: 0.97 | Min Dur: 18s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 25s | Thresh: 0.97 | Min Dur: 19s -> Caught: 19 | FA: 7 |
CSI: 0.6333
Window: 25s | Thresh: 0.97 | Min Dur: 20s -> Caught: 19 | FA: 7 |
CSI: 0.6333
Window: 25s | Thresh: 0.97 | Min Dur: 21s -> Caught: 19 | FA: 7 |
CSI: 0.6333
Window: 25s | Thresh: 0.97 | Min Dur: 22s -> Caught: 19 | FA: 7 |
CSI: 0.6333
Window: 25s | Thresh: 0.97 | Min Dur: 23s -> Caught: 19 | FA: 7 |
CSI: 0.6333
Window: 25s | Thresh: 0.97 | Min Dur: 24s -> Caught: 19 | FA: 7 |
CSI: 0.6333
Window: 25s | Thresh: 0.98 | Min Dur: 5s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 25s | Thresh: 0.98 | Min Dur: 6s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 25s | Thresh: 0.98 | Min Dur: 7s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 25s | Thresh: 0.98 | Min Dur: 8s -> Caught: 19 | FA: 8 |
CSI: 0.6129

```

Window: 25s		Thresh: 0.98		Min Dur: 9s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 25s		Thresh: 0.98		Min Dur: 10s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 25s		Thresh: 0.98		Min Dur: 11s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 25s		Thresh: 0.98		Min Dur: 12s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 25s		Thresh: 0.98		Min Dur: 13s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 25s		Thresh: 0.98		Min Dur: 14s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 25s		Thresh: 0.98		Min Dur: 15s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 25s		Thresh: 0.98		Min Dur: 16s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 25s		Thresh: 0.98		Min Dur: 17s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 25s		Thresh: 0.98		Min Dur: 18s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 25s		Thresh: 0.98		Min Dur: 19s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 25s		Thresh: 0.98		Min Dur: 20s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 25s		Thresh: 0.98		Min Dur: 21s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 25s		Thresh: 0.98		Min Dur: 22s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 25s		Thresh: 0.98		Min Dur: 23s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 25s		Thresh: 0.98		Min Dur: 24s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 25s		Thresh: 0.99		Min Dur: 5s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 25s		Thresh: 0.99		Min Dur: 6s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 25s		Thresh: 0.99		Min Dur: 7s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 25s		Thresh: 0.99		Min Dur: 8s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 25s		Thresh: 0.99		Min Dur: 9s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 25s		Thresh: 0.99		Min Dur: 10s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 25s		Thresh: 0.99		Min Dur: 11s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 25s		Thresh: 0.99		Min Dur: 12s	->	Caught: 15		FA: 8	
CSI: 0.4839									
Window: 25s		Thresh: 0.99		Min Dur: 13s	->	Caught: 15		FA: 8	

```

CSI: 0.4839
Window: 25s | Thresh: 0.99 | Min Dur: 14s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 25s | Thresh: 0.99 | Min Dur: 15s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 25s | Thresh: 0.99 | Min Dur: 16s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 25s | Thresh: 0.99 | Min Dur: 17s -> Caught: 15 | FA: 6 |
CSI: 0.5172
Window: 25s | Thresh: 0.99 | Min Dur: 18s -> Caught: 15 | FA: 6 |
CSI: 0.5172
Window: 25s | Thresh: 0.99 | Min Dur: 19s -> Caught: 15 | FA: 6 |
CSI: 0.5172
Window: 25s | Thresh: 0.99 | Min Dur: 20s -> Caught: 15 | FA: 6 |
CSI: 0.5172
Window: 25s | Thresh: 0.99 | Min Dur: 21s -> Caught: 15 | FA: 6 |
CSI: 0.5172
Window: 25s | Thresh: 0.99 | Min Dur: 22s -> Caught: 15 | FA: 6 |
CSI: 0.5172
Window: 25s | Thresh: 0.99 | Min Dur: 23s -> Caught: 15 | FA: 6 |
CSI: 0.5172
Window: 25s | Thresh: 0.99 | Min Dur: 24s -> Caught: 15 | FA: 6 |
CSI: 0.5172
Window: 30s | Thresh: 0.97 | Min Dur: 5s -> Caught: 22 | FA: 12 |
CSI: 0.6286
Window: 30s | Thresh: 0.97 | Min Dur: 6s -> Caught: 22 | FA: 11 |
CSI: 0.6471
Window: 30s | Thresh: 0.97 | Min Dur: 7s -> Caught: 22 | FA: 11 |
CSI: 0.6471
Window: 30s | Thresh: 0.97 | Min Dur: 8s -> Caught: 22 | FA: 12 |
CSI: 0.6286
Window: 30s | Thresh: 0.97 | Min Dur: 9s -> Caught: 21 | FA: 12 |
CSI: 0.6000
Window: 30s | Thresh: 0.97 | Min Dur: 10s -> Caught: 20 | FA: 12 |
CSI: 0.5714
Window: 30s | Thresh: 0.97 | Min Dur: 11s -> Caught: 19 | FA: 12 |
CSI: 0.5429
Window: 30s | Thresh: 0.97 | Min Dur: 12s -> Caught: 19 | FA: 11 |
CSI: 0.5588
Window: 30s | Thresh: 0.97 | Min Dur: 13s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 30s | Thresh: 0.97 | Min Dur: 14s -> Caught: 19 | FA: 13 |
CSI: 0.5278
Window: 30s | Thresh: 0.97 | Min Dur: 15s -> Caught: 19 | FA: 11 |
CSI: 0.5588
Window: 30s | Thresh: 0.97 | Min Dur: 16s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 30s | Thresh: 0.97 | Min Dur: 17s -> Caught: 19 | FA: 10 |
CSI: 0.5758

```


Window: 30s		Thresh: 0.97		Min Dur: 18s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 30s		Thresh: 0.97		Min Dur: 19s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 30s		Thresh: 0.97		Min Dur: 20s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 30s		Thresh: 0.97		Min Dur: 21s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 30s		Thresh: 0.97		Min Dur: 22s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 30s		Thresh: 0.97		Min Dur: 23s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 30s		Thresh: 0.97		Min Dur: 24s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 30s		Thresh: 0.98		Min Dur: 5s	->	Caught: 19		FA: 11	
CSI: 0.5588									
Window: 30s		Thresh: 0.98		Min Dur: 6s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 30s		Thresh: 0.98		Min Dur: 7s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 30s		Thresh: 0.98		Min Dur: 8s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 30s		Thresh: 0.98		Min Dur: 9s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 30s		Thresh: 0.98		Min Dur: 10s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 30s		Thresh: 0.98		Min Dur: 11s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 30s		Thresh: 0.98		Min Dur: 12s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 30s		Thresh: 0.98		Min Dur: 13s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 30s		Thresh: 0.98		Min Dur: 14s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 30s		Thresh: 0.98		Min Dur: 15s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 30s		Thresh: 0.98		Min Dur: 16s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 30s		Thresh: 0.98		Min Dur: 17s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 30s		Thresh: 0.98		Min Dur: 18s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 30s		Thresh: 0.98		Min Dur: 19s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 30s		Thresh: 0.98		Min Dur: 20s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 30s		Thresh: 0.98		Min Dur: 21s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 30s		Thresh: 0.98		Min Dur: 22s	->	Caught: 19		FA: 7	

```

CSI: 0.6333
Window: 30s | Thresh: 0.98 | Min Dur: 23s -> Caught: 19 | FA: 7 |
CSI: 0.6333
Window: 30s | Thresh: 0.98 | Min Dur: 24s -> Caught: 19 | FA: 7 |
CSI: 0.6333
Window: 30s | Thresh: 0.99 | Min Dur: 5s -> Caught: 15 | FA: 9 |
CSI: 0.4688
Window: 30s | Thresh: 0.99 | Min Dur: 6s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 30s | Thresh: 0.99 | Min Dur: 7s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 30s | Thresh: 0.99 | Min Dur: 8s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 30s | Thresh: 0.99 | Min Dur: 9s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 30s | Thresh: 0.99 | Min Dur: 10s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 30s | Thresh: 0.99 | Min Dur: 11s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 30s | Thresh: 0.99 | Min Dur: 12s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 30s | Thresh: 0.99 | Min Dur: 13s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 30s | Thresh: 0.99 | Min Dur: 14s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 30s | Thresh: 0.99 | Min Dur: 15s -> Caught: 15 | FA: 8 |
CSI: 0.4839
Window: 30s | Thresh: 0.99 | Min Dur: 16s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 30s | Thresh: 0.99 | Min Dur: 17s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 30s | Thresh: 0.99 | Min Dur: 18s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 30s | Thresh: 0.99 | Min Dur: 19s -> Caught: 15 | FA: 6 |
CSI: 0.5172
Window: 30s | Thresh: 0.99 | Min Dur: 20s -> Caught: 15 | FA: 6 |
CSI: 0.5172
Window: 30s | Thresh: 0.99 | Min Dur: 21s -> Caught: 15 | FA: 6 |
CSI: 0.5172
Window: 30s | Thresh: 0.99 | Min Dur: 22s -> Caught: 15 | FA: 6 |
CSI: 0.5172
Window: 30s | Thresh: 0.99 | Min Dur: 23s -> Caught: 15 | FA: 6 |
CSI: 0.5172
Window: 30s | Thresh: 0.99 | Min Dur: 24s -> Caught: 15 | FA: 6 |
CSI: 0.5172
Window: 35s | Thresh: 0.97 | Min Dur: 5s -> Caught: 22 | FA: 12 |
CSI: 0.6286
Window: 35s | Thresh: 0.97 | Min Dur: 6s -> Caught: 22 | FA: 11 |
CSI: 0.6471

```

Window: 35s	Thresh: 0.97	Min Dur: 7s	-> Caught: 22	FA: 10
CSI: 0.6667				
Window: 35s	Thresh: 0.97	Min Dur: 8s	-> Caught: 22	FA: 11
CSI: 0.6471				
Window: 35s	Thresh: 0.97	Min Dur: 9s	-> Caught: 21	FA: 11
CSI: 0.6176				
Window: 35s	Thresh: 0.97	Min Dur: 10s	-> Caught: 20	FA: 12
CSI: 0.5714				
Window: 35s	Thresh: 0.97	Min Dur: 11s	-> Caught: 19	FA: 12
CSI: 0.5429				
Window: 35s	Thresh: 0.97	Min Dur: 12s	-> Caught: 19	FA: 11
CSI: 0.5588				
Window: 35s	Thresh: 0.97	Min Dur: 13s	-> Caught: 19	FA: 11
CSI: 0.5588				
Window: 35s	Thresh: 0.97	Min Dur: 14s	-> Caught: 19	FA: 11
CSI: 0.5588				
Window: 35s	Thresh: 0.97	Min Dur: 15s	-> Caught: 19	FA: 11
CSI: 0.5588				
Window: 35s	Thresh: 0.97	Min Dur: 16s	-> Caught: 19	FA: 11
CSI: 0.5588				
Window: 35s	Thresh: 0.97	Min Dur: 17s	-> Caught: 19	FA: 12
CSI: 0.5429				
Window: 35s	Thresh: 0.97	Min Dur: 18s	-> Caught: 19	FA: 10
CSI: 0.5758				
Window: 35s	Thresh: 0.97	Min Dur: 19s	-> Caught: 19	FA: 10
CSI: 0.5758				
Window: 35s	Thresh: 0.97	Min Dur: 20s	-> Caught: 19	FA: 8
CSI: 0.6129				
Window: 35s	Thresh: 0.97	Min Dur: 21s	-> Caught: 19	FA: 8
CSI: 0.6129				
Window: 35s	Thresh: 0.97	Min Dur: 22s	-> Caught: 19	FA: 8
CSI: 0.6129				
Window: 35s	Thresh: 0.97	Min Dur: 23s	-> Caught: 19	FA: 7
CSI: 0.6333				
Window: 35s	Thresh: 0.97	Min Dur: 24s	-> Caught: 19	FA: 7
CSI: 0.6333				
Window: 35s	Thresh: 0.98	Min Dur: 5s	-> Caught: 19	FA: 12
CSI: 0.5429				
Window: 35s	Thresh: 0.98	Min Dur: 6s	-> Caught: 19	FA: 10
CSI: 0.5758				
Window: 35s	Thresh: 0.98	Min Dur: 7s	-> Caught: 19	FA: 9
CSI: 0.5938				
Window: 35s	Thresh: 0.98	Min Dur: 8s	-> Caught: 19	FA: 8
CSI: 0.6129				
Window: 35s	Thresh: 0.98	Min Dur: 9s	-> Caught: 19	FA: 8
CSI: 0.6129				
Window: 35s	Thresh: 0.98	Min Dur: 10s	-> Caught: 19	FA: 8
CSI: 0.6129				
Window: 35s	Thresh: 0.98	Min Dur: 11s	-> Caught: 19	FA: 8
CSI: 0.6129				

Window: 35s		Thresh: 0.98		Min Dur: 12s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 35s		Thresh: 0.98		Min Dur: 13s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 35s		Thresh: 0.98		Min Dur: 14s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 35s		Thresh: 0.98		Min Dur: 15s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 35s		Thresh: 0.98		Min Dur: 16s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 35s		Thresh: 0.98		Min Dur: 17s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 35s		Thresh: 0.98		Min Dur: 18s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 35s		Thresh: 0.98		Min Dur: 19s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 35s		Thresh: 0.98		Min Dur: 20s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 35s		Thresh: 0.98		Min Dur: 21s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 35s		Thresh: 0.98		Min Dur: 22s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 35s		Thresh: 0.98		Min Dur: 23s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 35s		Thresh: 0.98		Min Dur: 24s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 35s		Thresh: 0.99		Min Dur: 5s	->	Caught: 15		FA: 8	
CSI: 0.4839									
Window: 35s		Thresh: 0.99		Min Dur: 6s	->	Caught: 15		FA: 9	
CSI: 0.4688									
Window: 35s		Thresh: 0.99		Min Dur: 7s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 35s		Thresh: 0.99		Min Dur: 8s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 35s		Thresh: 0.99		Min Dur: 9s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 35s		Thresh: 0.99		Min Dur: 10s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 35s		Thresh: 0.99		Min Dur: 11s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 35s		Thresh: 0.99		Min Dur: 12s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 35s		Thresh: 0.99		Min Dur: 13s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 35s		Thresh: 0.99		Min Dur: 14s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 35s		Thresh: 0.99		Min Dur: 15s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 35s		Thresh: 0.99		Min Dur: 16s	->	Caught: 15		FA: 7	

CSI: 0.5000
Window: 35s Thresh: 0.99 Min Dur: 17s -> Caught: 15 FA: 8
CSI: 0.4839
Window: 35s Thresh: 0.99 Min Dur: 18s -> Caught: 15 FA: 7
CSI: 0.5000
Window: 35s Thresh: 0.99 Min Dur: 19s -> Caught: 15 FA: 7
CSI: 0.5000
Window: 35s Thresh: 0.99 Min Dur: 20s -> Caught: 15 FA: 7
CSI: 0.5000
Window: 35s Thresh: 0.99 Min Dur: 21s -> Caught: 15 FA: 7
CSI: 0.5000
Window: 35s Thresh: 0.99 Min Dur: 22s -> Caught: 15 FA: 6
CSI: 0.5172
Window: 35s Thresh: 0.99 Min Dur: 23s -> Caught: 15 FA: 6
CSI: 0.5172
Window: 35s Thresh: 0.99 Min Dur: 24s -> Caught: 15 FA: 6
CSI: 0.5172
Window: 40s Thresh: 0.97 Min Dur: 5s -> Caught: 22 FA: 12
CSI: 0.6286
Window: 40s Thresh: 0.97 Min Dur: 6s -> Caught: 22 FA: 12
CSI: 0.6286
Window: 40s Thresh: 0.97 Min Dur: 7s -> Caught: 22 FA: 10
CSI: 0.6667
Window: 40s Thresh: 0.97 Min Dur: 8s -> Caught: 22 FA: 11
CSI: 0.6471
Window: 40s Thresh: 0.97 Min Dur: 9s -> Caught: 21 FA: 11
CSI: 0.6176
Window: 40s Thresh: 0.97 Min Dur: 10s -> Caught: 20 FA: 12
CSI: 0.5714
Window: 40s Thresh: 0.97 Min Dur: 11s -> Caught: 19 FA: 12
CSI: 0.5429
Window: 40s Thresh: 0.97 Min Dur: 12s -> Caught: 19 FA: 11
CSI: 0.5588
Window: 40s Thresh: 0.97 Min Dur: 13s -> Caught: 19 FA: 11
CSI: 0.5588
Window: 40s Thresh: 0.97 Min Dur: 14s -> Caught: 19 FA: 11
CSI: 0.5588
Window: 40s Thresh: 0.97 Min Dur: 15s -> Caught: 19 FA: 11
CSI: 0.5588
Window: 40s Thresh: 0.97 Min Dur: 16s -> Caught: 19 FA: 11
CSI: 0.5588
Window: 40s Thresh: 0.97 Min Dur: 17s -> Caught: 19 FA: 10
CSI: 0.5758
Window: 40s Thresh: 0.97 Min Dur: 18s -> Caught: 19 FA: 10
CSI: 0.5758
Window: 40s Thresh: 0.97 Min Dur: 19s -> Caught: 19 FA: 11
CSI: 0.5588
Window: 40s Thresh: 0.97 Min Dur: 20s -> Caught: 19 FA: 10
CSI: 0.5758

Window: 40s		Thresh: 0.97		Min Dur: 21s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 40s		Thresh: 0.97		Min Dur: 22s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 40s		Thresh: 0.97		Min Dur: 23s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 40s		Thresh: 0.97		Min Dur: 24s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 40s		Thresh: 0.98		Min Dur: 5s	->	Caught: 19		FA: 12	
CSI: 0.5429									
Window: 40s		Thresh: 0.98		Min Dur: 6s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 40s		Thresh: 0.98		Min Dur: 7s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 40s		Thresh: 0.98		Min Dur: 8s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 40s		Thresh: 0.98		Min Dur: 9s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 40s		Thresh: 0.98		Min Dur: 10s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 40s		Thresh: 0.98		Min Dur: 11s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 40s		Thresh: 0.98		Min Dur: 12s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 40s		Thresh: 0.98		Min Dur: 13s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 40s		Thresh: 0.98		Min Dur: 14s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 40s		Thresh: 0.98		Min Dur: 15s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 40s		Thresh: 0.98		Min Dur: 16s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 40s		Thresh: 0.98		Min Dur: 17s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 40s		Thresh: 0.98		Min Dur: 18s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 40s		Thresh: 0.98		Min Dur: 19s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 40s		Thresh: 0.98		Min Dur: 20s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 40s		Thresh: 0.98		Min Dur: 21s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 40s		Thresh: 0.98		Min Dur: 22s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 40s		Thresh: 0.98		Min Dur: 23s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 40s		Thresh: 0.98		Min Dur: 24s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 40s		Thresh: 0.99		Min Dur: 5s	->	Caught: 15		FA: 8	

```

CSI: 0.4839
Window: 40s | Thresh: 0.99 | Min Dur: 6s -> Caught: 15 | FA: 9 |
CSI: 0.4688
Window: 40s | Thresh: 0.99 | Min Dur: 7s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 40s | Thresh: 0.99 | Min Dur: 8s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 40s | Thresh: 0.99 | Min Dur: 9s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 40s | Thresh: 0.99 | Min Dur: 10s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 40s | Thresh: 0.99 | Min Dur: 11s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 40s | Thresh: 0.99 | Min Dur: 12s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 40s | Thresh: 0.99 | Min Dur: 13s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 40s | Thresh: 0.99 | Min Dur: 14s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 40s | Thresh: 0.99 | Min Dur: 15s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 40s | Thresh: 0.99 | Min Dur: 16s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 40s | Thresh: 0.99 | Min Dur: 17s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 40s | Thresh: 0.99 | Min Dur: 18s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 40s | Thresh: 0.99 | Min Dur: 19s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 40s | Thresh: 0.99 | Min Dur: 20s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 40s | Thresh: 0.99 | Min Dur: 21s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 40s | Thresh: 0.99 | Min Dur: 22s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 40s | Thresh: 0.99 | Min Dur: 23s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 40s | Thresh: 0.99 | Min Dur: 24s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 45s | Thresh: 0.97 | Min Dur: 5s -> Caught: 22 | FA: 12 |
CSI: 0.6286
Window: 45s | Thresh: 0.97 | Min Dur: 6s -> Caught: 22 | FA: 12 |
CSI: 0.6286
Window: 45s | Thresh: 0.97 | Min Dur: 7s -> Caught: 22 | FA: 11 |
CSI: 0.6471
Window: 45s | Thresh: 0.97 | Min Dur: 8s -> Caught: 22 | FA: 11 |
CSI: 0.6471
Window: 45s | Thresh: 0.97 | Min Dur: 9s -> Caught: 21 | FA: 12 |
CSI: 0.6000

```

Window: 45s		Thresh: 0.97		Min Dur: 10s	->	Caught: 20		FA: 11	
CSI: 0.5882									
Window: 45s		Thresh: 0.97		Min Dur: 11s	->	Caught: 19		FA: 11	
CSI: 0.5588									
Window: 45s		Thresh: 0.97		Min Dur: 12s	->	Caught: 19		FA: 12	
CSI: 0.5429									
Window: 45s		Thresh: 0.97		Min Dur: 13s	->	Caught: 19		FA: 11	
CSI: 0.5588									
Window: 45s		Thresh: 0.97		Min Dur: 14s	->	Caught: 19		FA: 11	
CSI: 0.5588									
Window: 45s		Thresh: 0.97		Min Dur: 15s	->	Caught: 19		FA: 11	
CSI: 0.5588									
Window: 45s		Thresh: 0.97		Min Dur: 16s	->	Caught: 19		FA: 11	
CSI: 0.5588									
Window: 45s		Thresh: 0.97		Min Dur: 17s	->	Caught: 19		FA: 11	
CSI: 0.5588									
Window: 45s		Thresh: 0.97		Min Dur: 18s	->	Caught: 19		FA: 11	
CSI: 0.5588									
Window: 45s		Thresh: 0.97		Min Dur: 19s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 45s		Thresh: 0.97		Min Dur: 20s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 45s		Thresh: 0.97		Min Dur: 21s	->	Caught: 19		FA: 11	
CSI: 0.5588									
Window: 45s		Thresh: 0.97		Min Dur: 22s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 45s		Thresh: 0.97		Min Dur: 23s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 45s		Thresh: 0.97		Min Dur: 24s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 45s		Thresh: 0.98		Min Dur: 5s	->	Caught: 19		FA: 12	
CSI: 0.5429									
Window: 45s		Thresh: 0.98		Min Dur: 6s	->	Caught: 19		FA: 11	
CSI: 0.5588									
Window: 45s		Thresh: 0.98		Min Dur: 7s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 45s		Thresh: 0.98		Min Dur: 8s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 45s		Thresh: 0.98		Min Dur: 9s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 45s		Thresh: 0.98		Min Dur: 10s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 45s		Thresh: 0.98		Min Dur: 11s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 45s		Thresh: 0.98		Min Dur: 12s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 45s		Thresh: 0.98		Min Dur: 13s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 45s		Thresh: 0.98		Min Dur: 14s	->	Caught: 19		FA: 8	

CSI: 0.6129					
Window: 45s		Thresh: 0.98		Min Dur: 15s ->	Caught: 19 FA: 8
CSI: 0.6129					
Window: 45s		Thresh: 0.98		Min Dur: 16s ->	Caught: 19 FA: 8
CSI: 0.6129					
Window: 45s		Thresh: 0.98		Min Dur: 17s ->	Caught: 19 FA: 7
CSI: 0.6333					
Window: 45s		Thresh: 0.98		Min Dur: 18s ->	Caught: 19 FA: 7
CSI: 0.6333					
Window: 45s		Thresh: 0.98		Min Dur: 19s ->	Caught: 19 FA: 7
CSI: 0.6333					
Window: 45s		Thresh: 0.98		Min Dur: 20s ->	Caught: 19 FA: 7
CSI: 0.6333					
Window: 45s		Thresh: 0.98		Min Dur: 21s ->	Caught: 19 FA: 7
CSI: 0.6333					
Window: 45s		Thresh: 0.98		Min Dur: 22s ->	Caught: 19 FA: 7
CSI: 0.6333					
Window: 45s		Thresh: 0.98		Min Dur: 23s ->	Caught: 19 FA: 7
CSI: 0.6333					
Window: 45s		Thresh: 0.98		Min Dur: 24s ->	Caught: 19 FA: 7
CSI: 0.6333					
Window: 45s		Thresh: 0.99		Min Dur: 5s ->	Caught: 15 FA: 8
CSI: 0.4839					
Window: 45s		Thresh: 0.99		Min Dur: 6s ->	Caught: 15 FA: 8
CSI: 0.4839					
Window: 45s		Thresh: 0.99		Min Dur: 7s ->	Caught: 15 FA: 8
CSI: 0.4839					
Window: 45s		Thresh: 0.99		Min Dur: 8s ->	Caught: 15 FA: 7
CSI: 0.5000					
Window: 45s		Thresh: 0.99		Min Dur: 9s ->	Caught: 15 FA: 7
CSI: 0.5000					
Window: 45s		Thresh: 0.99		Min Dur: 10s ->	Caught: 15 FA: 7
CSI: 0.5000					
Window: 45s		Thresh: 0.99		Min Dur: 11s ->	Caught: 15 FA: 7
CSI: 0.5000					
Window: 45s		Thresh: 0.99		Min Dur: 12s ->	Caught: 15 FA: 7
CSI: 0.5000					
Window: 45s		Thresh: 0.99		Min Dur: 13s ->	Caught: 15 FA: 7
CSI: 0.5000					
Window: 45s		Thresh: 0.99		Min Dur: 14s ->	Caught: 15 FA: 7
CSI: 0.5000					
Window: 45s		Thresh: 0.99		Min Dur: 15s ->	Caught: 15 FA: 7
CSI: 0.5000					
Window: 45s		Thresh: 0.99		Min Dur: 16s ->	Caught: 15 FA: 7
CSI: 0.5000					
Window: 45s		Thresh: 0.99		Min Dur: 17s ->	Caught: 15 FA: 7
CSI: 0.5000					
Window: 45s		Thresh: 0.99		Min Dur: 18s ->	Caught: 15 FA: 7
CSI: 0.5000					

Window: 45s	Thresh: 0.99	Min Dur: 19s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 45s	Thresh: 0.99	Min Dur: 20s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 45s	Thresh: 0.99	Min Dur: 21s	-> Caught: 15	FA: 8
CSI: 0.4839				
Window: 45s	Thresh: 0.99	Min Dur: 22s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 45s	Thresh: 0.99	Min Dur: 23s	-> Caught: 15	FA: 8
CSI: 0.4839				
Window: 45s	Thresh: 0.99	Min Dur: 24s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 50s	Thresh: 0.97	Min Dur: 5s	-> Caught: 22	FA: 12
CSI: 0.6286				
Window: 50s	Thresh: 0.97	Min Dur: 6s	-> Caught: 22	FA: 12
CSI: 0.6286				
Window: 50s	Thresh: 0.97	Min Dur: 7s	-> Caught: 22	FA: 11
CSI: 0.6471				
Window: 50s	Thresh: 0.97	Min Dur: 8s	-> Caught: 22	FA: 11
CSI: 0.6471				
Window: 50s	Thresh: 0.97	Min Dur: 9s	-> Caught: 21	FA: 12
CSI: 0.6000				
Window: 50s	Thresh: 0.97	Min Dur: 10s	-> Caught: 20	FA: 12
CSI: 0.5714				
Window: 50s	Thresh: 0.97	Min Dur: 11s	-> Caught: 19	FA: 11
CSI: 0.5588				
Window: 50s	Thresh: 0.97	Min Dur: 12s	-> Caught: 19	FA: 11
CSI: 0.5588				
Window: 50s	Thresh: 0.97	Min Dur: 13s	-> Caught: 19	FA: 12
CSI: 0.5429				
Window: 50s	Thresh: 0.97	Min Dur: 14s	-> Caught: 19	FA: 11
CSI: 0.5588				
Window: 50s	Thresh: 0.97	Min Dur: 15s	-> Caught: 19	FA: 11
CSI: 0.5588				
Window: 50s	Thresh: 0.97	Min Dur: 16s	-> Caught: 19	FA: 11
CSI: 0.5588				
Window: 50s	Thresh: 0.97	Min Dur: 17s	-> Caught: 19	FA: 11
CSI: 0.5588				
Window: 50s	Thresh: 0.97	Min Dur: 18s	-> Caught: 19	FA: 11
CSI: 0.5588				
Window: 50s	Thresh: 0.97	Min Dur: 19s	-> Caught: 19	FA: 11
CSI: 0.5588				
Window: 50s	Thresh: 0.97	Min Dur: 20s	-> Caught: 19	FA: 11
CSI: 0.5588				
Window: 50s	Thresh: 0.97	Min Dur: 21s	-> Caught: 19	FA: 9
CSI: 0.5938				
Window: 50s	Thresh: 0.97	Min Dur: 22s	-> Caught: 19	FA: 11
CSI: 0.5588				
Window: 50s	Thresh: 0.97	Min Dur: 23s	-> Caught: 19	FA: 11

CSI: 0.5588					
Window: 50s		Thresh: 0.97		Min Dur: 24s ->	Caught: 19 FA: 9
CSI: 0.5938					
Window: 50s		Thresh: 0.98		Min Dur: 5s ->	Caught: 19 FA: 12
CSI: 0.5429					
Window: 50s		Thresh: 0.98		Min Dur: 6s ->	Caught: 19 FA: 12
CSI: 0.5429					
Window: 50s		Thresh: 0.98		Min Dur: 7s ->	Caught: 19 FA: 9
CSI: 0.5938					
Window: 50s		Thresh: 0.98		Min Dur: 8s ->	Caught: 19 FA: 9
CSI: 0.5938					
Window: 50s		Thresh: 0.98		Min Dur: 9s ->	Caught: 19 FA: 9
CSI: 0.5938					
Window: 50s		Thresh: 0.98		Min Dur: 10s ->	Caught: 19 FA: 8
CSI: 0.6129					
Window: 50s		Thresh: 0.98		Min Dur: 11s ->	Caught: 19 FA: 8
CSI: 0.6129					
Window: 50s		Thresh: 0.98		Min Dur: 12s ->	Caught: 19 FA: 8
CSI: 0.6129					
Window: 50s		Thresh: 0.98		Min Dur: 13s ->	Caught: 19 FA: 8
CSI: 0.6129					
Window: 50s		Thresh: 0.98		Min Dur: 14s ->	Caught: 19 FA: 8
CSI: 0.6129					
Window: 50s		Thresh: 0.98		Min Dur: 15s ->	Caught: 19 FA: 8
CSI: 0.6129					
Window: 50s		Thresh: 0.98		Min Dur: 16s ->	Caught: 19 FA: 8
CSI: 0.6129					
Window: 50s		Thresh: 0.98		Min Dur: 17s ->	Caught: 19 FA: 8
CSI: 0.6129					
Window: 50s		Thresh: 0.98		Min Dur: 18s ->	Caught: 19 FA: 8
CSI: 0.6129					
Window: 50s		Thresh: 0.98		Min Dur: 19s ->	Caught: 19 FA: 7
CSI: 0.6333					
Window: 50s		Thresh: 0.98		Min Dur: 20s ->	Caught: 19 FA: 7
CSI: 0.6333					
Window: 50s		Thresh: 0.98		Min Dur: 21s ->	Caught: 19 FA: 7
CSI: 0.6333					
Window: 50s		Thresh: 0.98		Min Dur: 22s ->	Caught: 19 FA: 7
CSI: 0.6333					
Window: 50s		Thresh: 0.98		Min Dur: 23s ->	Caught: 19 FA: 7
CSI: 0.6333					
Window: 50s		Thresh: 0.98		Min Dur: 24s ->	Caught: 19 FA: 7
CSI: 0.6333					
Window: 50s		Thresh: 0.99		Min Dur: 5s ->	Caught: 15 FA: 8
CSI: 0.4839					
Window: 50s		Thresh: 0.99		Min Dur: 6s ->	Caught: 15 FA: 8
CSI: 0.4839					
Window: 50s		Thresh: 0.99		Min Dur: 7s ->	Caught: 15 FA: 9
CSI: 0.4688					

Window: 50s		Thresh: 0.99		Min Dur: 8s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 50s		Thresh: 0.99		Min Dur: 9s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 50s		Thresh: 0.99		Min Dur: 10s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 50s		Thresh: 0.99		Min Dur: 11s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 50s		Thresh: 0.99		Min Dur: 12s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 50s		Thresh: 0.99		Min Dur: 13s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 50s		Thresh: 0.99		Min Dur: 14s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 50s		Thresh: 0.99		Min Dur: 15s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 50s		Thresh: 0.99		Min Dur: 16s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 50s		Thresh: 0.99		Min Dur: 17s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 50s		Thresh: 0.99		Min Dur: 18s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 50s		Thresh: 0.99		Min Dur: 19s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 50s		Thresh: 0.99		Min Dur: 20s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 50s		Thresh: 0.99		Min Dur: 21s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 50s		Thresh: 0.99		Min Dur: 22s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 50s		Thresh: 0.99		Min Dur: 23s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 50s		Thresh: 0.99		Min Dur: 24s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 55s		Thresh: 0.97		Min Dur: 5s	->	Caught: 22		FA: 13	
CSI: 0.6111									
Window: 55s		Thresh: 0.97		Min Dur: 6s	->	Caught: 22		FA: 12	
CSI: 0.6286									
Window: 55s		Thresh: 0.97		Min Dur: 7s	->	Caught: 22		FA: 11	
CSI: 0.6471									
Window: 55s		Thresh: 0.97		Min Dur: 8s	->	Caught: 22		FA: 11	
CSI: 0.6471									
Window: 55s		Thresh: 0.97		Min Dur: 9s	->	Caught: 21		FA: 11	
CSI: 0.6176									
Window: 55s		Thresh: 0.97		Min Dur: 10s	->	Caught: 20		FA: 12	
CSI: 0.5714									
Window: 55s		Thresh: 0.97		Min Dur: 11s	->	Caught: 19		FA: 12	
CSI: 0.5429									
Window: 55s		Thresh: 0.97		Min Dur: 12s	->	Caught: 19		FA: 11	

CSI: 0.5588					
Window: 55s		Thresh: 0.97		Min Dur: 13s ->	Caught: 19 FA: 11
CSI: 0.5588					
Window: 55s		Thresh: 0.97		Min Dur: 14s ->	Caught: 19 FA: 10
CSI: 0.5758					
Window: 55s		Thresh: 0.97		Min Dur: 15s ->	Caught: 19 FA: 11
CSI: 0.5588					
Window: 55s		Thresh: 0.97		Min Dur: 16s ->	Caught: 19 FA: 11
CSI: 0.5588					
Window: 55s		Thresh: 0.97		Min Dur: 17s ->	Caught: 19 FA: 11
CSI: 0.5588					
Window: 55s		Thresh: 0.97		Min Dur: 18s ->	Caught: 19 FA: 11
CSI: 0.5588					
Window: 55s		Thresh: 0.97		Min Dur: 19s ->	Caught: 19 FA: 11
CSI: 0.5588					
Window: 55s		Thresh: 0.97		Min Dur: 20s ->	Caught: 19 FA: 11
CSI: 0.5588					
Window: 55s		Thresh: 0.97		Min Dur: 21s ->	Caught: 19 FA: 11
CSI: 0.5588					
Window: 55s		Thresh: 0.97		Min Dur: 22s ->	Caught: 19 FA: 11
CSI: 0.5588					
Window: 55s		Thresh: 0.97		Min Dur: 23s ->	Caught: 19 FA: 10
CSI: 0.5758					
Window: 55s		Thresh: 0.97		Min Dur: 24s ->	Caught: 19 FA: 10
CSI: 0.5758					
Window: 55s		Thresh: 0.98		Min Dur: 5s ->	Caught: 19 FA: 12
CSI: 0.5429					
Window: 55s		Thresh: 0.98		Min Dur: 6s ->	Caught: 19 FA: 12
CSI: 0.5429					
Window: 55s		Thresh: 0.98		Min Dur: 7s ->	Caught: 19 FA: 10
CSI: 0.5758					
Window: 55s		Thresh: 0.98		Min Dur: 8s ->	Caught: 19 FA: 9
CSI: 0.5938					
Window: 55s		Thresh: 0.98		Min Dur: 9s ->	Caught: 19 FA: 10
CSI: 0.5758					
Window: 55s		Thresh: 0.98		Min Dur: 10s ->	Caught: 19 FA: 9
CSI: 0.5938					
Window: 55s		Thresh: 0.98		Min Dur: 11s ->	Caught: 19 FA: 8
CSI: 0.6129					
Window: 55s		Thresh: 0.98		Min Dur: 12s ->	Caught: 19 FA: 8
CSI: 0.6129					
Window: 55s		Thresh: 0.98		Min Dur: 13s ->	Caught: 19 FA: 8
CSI: 0.6129					
Window: 55s		Thresh: 0.98		Min Dur: 14s ->	Caught: 19 FA: 8
CSI: 0.6129					
Window: 55s		Thresh: 0.98		Min Dur: 15s ->	Caught: 19 FA: 8
CSI: 0.6129					
Window: 55s		Thresh: 0.98		Min Dur: 16s ->	Caught: 19 FA: 8
CSI: 0.6129					

Window: 55s	Thresh: 0.98	Min Dur: 17s	-> Caught: 19	FA: 8
CSI: 0.6129				
Window: 55s	Thresh: 0.98	Min Dur: 18s	-> Caught: 19	FA: 8
CSI: 0.6129				
Window: 55s	Thresh: 0.98	Min Dur: 19s	-> Caught: 19	FA: 8
CSI: 0.6129				
Window: 55s	Thresh: 0.98	Min Dur: 20s	-> Caught: 19	FA: 7
CSI: 0.6333				
Window: 55s	Thresh: 0.98	Min Dur: 21s	-> Caught: 19	FA: 7
CSI: 0.6333				
Window: 55s	Thresh: 0.98	Min Dur: 22s	-> Caught: 19	FA: 7
CSI: 0.6333				
Window: 55s	Thresh: 0.98	Min Dur: 23s	-> Caught: 19	FA: 7
CSI: 0.6333				
Window: 55s	Thresh: 0.98	Min Dur: 24s	-> Caught: 19	FA: 7
CSI: 0.6333				
Window: 55s	Thresh: 0.99	Min Dur: 5s	-> Caught: 15	FA: 8
CSI: 0.4839				
Window: 55s	Thresh: 0.99	Min Dur: 6s	-> Caught: 15	FA: 8
CSI: 0.4839				
Window: 55s	Thresh: 0.99	Min Dur: 7s	-> Caught: 15	FA: 9
CSI: 0.4688				
Window: 55s	Thresh: 0.99	Min Dur: 8s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 55s	Thresh: 0.99	Min Dur: 9s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 55s	Thresh: 0.99	Min Dur: 10s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 55s	Thresh: 0.99	Min Dur: 11s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 55s	Thresh: 0.99	Min Dur: 12s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 55s	Thresh: 0.99	Min Dur: 13s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 55s	Thresh: 0.99	Min Dur: 14s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 55s	Thresh: 0.99	Min Dur: 15s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 55s	Thresh: 0.99	Min Dur: 16s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 55s	Thresh: 0.99	Min Dur: 17s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 55s	Thresh: 0.99	Min Dur: 18s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 55s	Thresh: 0.99	Min Dur: 19s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 55s	Thresh: 0.99	Min Dur: 20s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 55s	Thresh: 0.99	Min Dur: 21s	-> Caught: 15	FA: 7
CSI: 0.5000				

Window: 55s		Thresh: 0.99		Min Dur: 22s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 55s		Thresh: 0.99		Min Dur: 23s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 55s		Thresh: 0.99		Min Dur: 24s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 60s		Thresh: 0.97		Min Dur: 5s	->	Caught: 22		FA: 13	
CSI: 0.6111									
Window: 60s		Thresh: 0.97		Min Dur: 6s	->	Caught: 22		FA: 12	
CSI: 0.6286									
Window: 60s		Thresh: 0.97		Min Dur: 7s	->	Caught: 22		FA: 11	
CSI: 0.6471									
Window: 60s		Thresh: 0.97		Min Dur: 8s	->	Caught: 22		FA: 11	
CSI: 0.6471									
Window: 60s		Thresh: 0.97		Min Dur: 9s	->	Caught: 21		FA: 10	
CSI: 0.6364									
Window: 60s		Thresh: 0.97		Min Dur: 10s	->	Caught: 20		FA: 11	
CSI: 0.5882									
Window: 60s		Thresh: 0.97		Min Dur: 11s	->	Caught: 19		FA: 11	
CSI: 0.5588									
Window: 60s		Thresh: 0.97		Min Dur: 12s	->	Caught: 19		FA: 11	
CSI: 0.5588									
Window: 60s		Thresh: 0.97		Min Dur: 13s	->	Caught: 19		FA: 11	
CSI: 0.5588									
Window: 60s		Thresh: 0.97		Min Dur: 14s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 60s		Thresh: 0.97		Min Dur: 15s	->	Caught: 19		FA: 11	
CSI: 0.5588									
Window: 60s		Thresh: 0.97		Min Dur: 16s	->	Caught: 19		FA: 11	
CSI: 0.5588									
Window: 60s		Thresh: 0.97		Min Dur: 17s	->	Caught: 19		FA: 11	
CSI: 0.5588									
Window: 60s		Thresh: 0.97		Min Dur: 18s	->	Caught: 19		FA: 11	
CSI: 0.5588									
Window: 60s		Thresh: 0.97		Min Dur: 19s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 60s		Thresh: 0.97		Min Dur: 20s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 60s		Thresh: 0.97		Min Dur: 21s	->	Caught: 19		FA: 11	
CSI: 0.5588									
Window: 60s		Thresh: 0.97		Min Dur: 22s	->	Caught: 19		FA: 11	
CSI: 0.5588									
Window: 60s		Thresh: 0.97		Min Dur: 23s	->	Caught: 19		FA: 11	
CSI: 0.5588									
Window: 60s		Thresh: 0.97		Min Dur: 24s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 60s		Thresh: 0.98		Min Dur: 5s	->	Caught: 19		FA: 12	
CSI: 0.5429									
Window: 60s		Thresh: 0.98		Min Dur: 6s	->	Caught: 19		FA: 12	

```

CSI: 0.5429
Window: 60s | Thresh: 0.98 | Min Dur: 7s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 60s | Thresh: 0.98 | Min Dur: 8s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 60s | Thresh: 0.98 | Min Dur: 9s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 60s | Thresh: 0.98 | Min Dur: 10s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 60s | Thresh: 0.98 | Min Dur: 11s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 60s | Thresh: 0.98 | Min Dur: 12s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 60s | Thresh: 0.98 | Min Dur: 13s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 60s | Thresh: 0.98 | Min Dur: 14s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 60s | Thresh: 0.98 | Min Dur: 15s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 60s | Thresh: 0.98 | Min Dur: 16s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 60s | Thresh: 0.98 | Min Dur: 17s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 60s | Thresh: 0.98 | Min Dur: 18s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 60s | Thresh: 0.98 | Min Dur: 19s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 60s | Thresh: 0.98 | Min Dur: 20s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 60s | Thresh: 0.98 | Min Dur: 21s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 60s | Thresh: 0.98 | Min Dur: 22s -> Caught: 19 | FA: 7 |
CSI: 0.6333
Window: 60s | Thresh: 0.98 | Min Dur: 23s -> Caught: 19 | FA: 7 |
CSI: 0.6333
Window: 60s | Thresh: 0.98 | Min Dur: 24s -> Caught: 19 | FA: 7 |
CSI: 0.6333
Window: 60s | Thresh: 0.99 | Min Dur: 5s -> Caught: 15 | FA: 8 |
CSI: 0.4839
Window: 60s | Thresh: 0.99 | Min Dur: 6s -> Caught: 15 | FA: 8 |
CSI: 0.4839
Window: 60s | Thresh: 0.99 | Min Dur: 7s -> Caught: 15 | FA: 9 |
CSI: 0.4688
Window: 60s | Thresh: 0.99 | Min Dur: 8s -> Caught: 15 | FA: 8 |
CSI: 0.4839
Window: 60s | Thresh: 0.99 | Min Dur: 9s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 60s | Thresh: 0.99 | Min Dur: 10s -> Caught: 15 | FA: 7 |
CSI: 0.5000

```


Window: 60s		Thresh: 0.99		Min Dur: 11s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 60s		Thresh: 0.99		Min Dur: 12s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 60s		Thresh: 0.99		Min Dur: 13s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 60s		Thresh: 0.99		Min Dur: 14s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 60s		Thresh: 0.99		Min Dur: 15s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 60s		Thresh: 0.99		Min Dur: 16s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 60s		Thresh: 0.99		Min Dur: 17s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 60s		Thresh: 0.99		Min Dur: 18s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 60s		Thresh: 0.99		Min Dur: 19s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 60s		Thresh: 0.99		Min Dur: 20s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 60s		Thresh: 0.99		Min Dur: 21s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 60s		Thresh: 0.99		Min Dur: 22s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 60s		Thresh: 0.99		Min Dur: 23s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 60s		Thresh: 0.99		Min Dur: 24s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 65s		Thresh: 0.97		Min Dur: 5s	->	Caught: 22		FA: 13	
CSI: 0.6111									
Window: 65s		Thresh: 0.97		Min Dur: 6s	->	Caught: 22		FA: 12	
CSI: 0.6286									
Window: 65s		Thresh: 0.97		Min Dur: 7s	->	Caught: 22		FA: 11	
CSI: 0.6471									
Window: 65s		Thresh: 0.97		Min Dur: 8s	->	Caught: 22		FA: 11	
CSI: 0.6471									
Window: 65s		Thresh: 0.97		Min Dur: 9s	->	Caught: 21		FA: 11	
CSI: 0.6176									
Window: 65s		Thresh: 0.97		Min Dur: 10s	->	Caught: 20		FA: 10	
CSI: 0.6061									
Window: 65s		Thresh: 0.97		Min Dur: 11s	->	Caught: 19		FA: 11	
CSI: 0.5588									
Window: 65s		Thresh: 0.97		Min Dur: 12s	->	Caught: 19		FA: 11	
CSI: 0.5588									
Window: 65s		Thresh: 0.97		Min Dur: 13s	->	Caught: 19		FA: 11	
CSI: 0.5588									
Window: 65s		Thresh: 0.97		Min Dur: 14s	->	Caught: 19		FA: 11	
CSI: 0.5588									
Window: 65s		Thresh: 0.97		Min Dur: 15s	->	Caught: 19		FA: 10	

CSI: 0.5758					
Window: 65s		Thresh: 0.97		Min Dur: 16s	-> Caught: 19 FA: 11
CSI: 0.5588					
Window: 65s		Thresh: 0.97		Min Dur: 17s	-> Caught: 19 FA: 11
CSI: 0.5588					
Window: 65s		Thresh: 0.97		Min Dur: 18s	-> Caught: 19 FA: 11
CSI: 0.5588					
Window: 65s		Thresh: 0.97		Min Dur: 19s	-> Caught: 19 FA: 10
CSI: 0.5758					
Window: 65s		Thresh: 0.97		Min Dur: 20s	-> Caught: 19 FA: 10
CSI: 0.5758					
Window: 65s		Thresh: 0.97		Min Dur: 21s	-> Caught: 19 FA: 10
CSI: 0.5758					
Window: 65s		Thresh: 0.97		Min Dur: 22s	-> Caught: 19 FA: 10
CSI: 0.5758					
Window: 65s		Thresh: 0.97		Min Dur: 23s	-> Caught: 19 FA: 11
CSI: 0.5588					
Window: 65s		Thresh: 0.97		Min Dur: 24s	-> Caught: 19 FA: 11
CSI: 0.5588					
Window: 65s		Thresh: 0.98		Min Dur: 5s	-> Caught: 19 FA: 12
CSI: 0.5429					
Window: 65s		Thresh: 0.98		Min Dur: 6s	-> Caught: 19 FA: 12
CSI: 0.5429					
Window: 65s		Thresh: 0.98		Min Dur: 7s	-> Caught: 19 FA: 10
CSI: 0.5758					
Window: 65s		Thresh: 0.98		Min Dur: 8s	-> Caught: 19 FA: 10
CSI: 0.5758					
Window: 65s		Thresh: 0.98		Min Dur: 9s	-> Caught: 19 FA: 9
CSI: 0.5938					
Window: 65s		Thresh: 0.98		Min Dur: 10s	-> Caught: 19 FA: 8
CSI: 0.6129					
Window: 65s		Thresh: 0.98		Min Dur: 11s	-> Caught: 19 FA: 8
CSI: 0.6129					
Window: 65s		Thresh: 0.98		Min Dur: 12s	-> Caught: 19 FA: 8
CSI: 0.6129					
Window: 65s		Thresh: 0.98		Min Dur: 13s	-> Caught: 19 FA: 8
CSI: 0.6129					
Window: 65s		Thresh: 0.98		Min Dur: 14s	-> Caught: 19 FA: 8
CSI: 0.6129					
Window: 65s		Thresh: 0.98		Min Dur: 15s	-> Caught: 19 FA: 8
CSI: 0.6129					
Window: 65s		Thresh: 0.98		Min Dur: 16s	-> Caught: 19 FA: 8
CSI: 0.6129					
Window: 65s		Thresh: 0.98		Min Dur: 17s	-> Caught: 19 FA: 8
CSI: 0.6129					
Window: 65s		Thresh: 0.98		Min Dur: 18s	-> Caught: 19 FA: 8
CSI: 0.6129					
Window: 65s		Thresh: 0.98		Min Dur: 19s	-> Caught: 19 FA: 8
CSI: 0.6129					

Window: 65s		Thresh: 0.98		Min Dur: 20s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 65s		Thresh: 0.98		Min Dur: 21s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 65s		Thresh: 0.98		Min Dur: 22s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 65s		Thresh: 0.98		Min Dur: 23s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 65s		Thresh: 0.98		Min Dur: 24s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 65s		Thresh: 0.99		Min Dur: 5s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 65s		Thresh: 0.99		Min Dur: 6s	->	Caught: 15		FA: 8	
CSI: 0.4839									
Window: 65s		Thresh: 0.99		Min Dur: 7s	->	Caught: 15		FA: 8	
CSI: 0.4839									
Window: 65s		Thresh: 0.99		Min Dur: 8s	->	Caught: 15		FA: 9	
CSI: 0.4688									
Window: 65s		Thresh: 0.99		Min Dur: 9s	->	Caught: 15		FA: 8	
CSI: 0.4839									
Window: 65s		Thresh: 0.99		Min Dur: 10s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 65s		Thresh: 0.99		Min Dur: 11s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 65s		Thresh: 0.99		Min Dur: 12s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 65s		Thresh: 0.99		Min Dur: 13s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 65s		Thresh: 0.99		Min Dur: 14s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 65s		Thresh: 0.99		Min Dur: 15s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 65s		Thresh: 0.99		Min Dur: 16s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 65s		Thresh: 0.99		Min Dur: 17s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 65s		Thresh: 0.99		Min Dur: 18s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 65s		Thresh: 0.99		Min Dur: 19s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 65s		Thresh: 0.99		Min Dur: 20s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 65s		Thresh: 0.99		Min Dur: 21s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 65s		Thresh: 0.99		Min Dur: 22s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 65s		Thresh: 0.99		Min Dur: 23s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 65s		Thresh: 0.99		Min Dur: 24s	->	Caught: 15		FA: 7	

```

CSI: 0.5000
Window: 70s | Thresh: 0.97 | Min Dur: 5s -> Caught: 22 | FA: 13 |
CSI: 0.6111
Window: 70s | Thresh: 0.97 | Min Dur: 6s -> Caught: 22 | FA: 12 |
CSI: 0.6286
Window: 70s | Thresh: 0.97 | Min Dur: 7s -> Caught: 22 | FA: 11 |
CSI: 0.6471
Window: 70s | Thresh: 0.97 | Min Dur: 8s -> Caught: 22 | FA: 11 |
CSI: 0.6471
Window: 70s | Thresh: 0.97 | Min Dur: 9s -> Caught: 21 | FA: 11 |
CSI: 0.6176
Window: 70s | Thresh: 0.97 | Min Dur: 10s -> Caught: 20 | FA: 10 |
CSI: 0.6061
Window: 70s | Thresh: 0.97 | Min Dur: 11s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 70s | Thresh: 0.97 | Min Dur: 12s -> Caught: 19 | FA: 11 |
CSI: 0.5588
Window: 70s | Thresh: 0.97 | Min Dur: 13s -> Caught: 19 | FA: 11 |
CSI: 0.5588
Window: 70s | Thresh: 0.97 | Min Dur: 14s -> Caught: 19 | FA: 11 |
CSI: 0.5588
Window: 70s | Thresh: 0.97 | Min Dur: 15s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 70s | Thresh: 0.97 | Min Dur: 16s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 70s | Thresh: 0.97 | Min Dur: 17s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 70s | Thresh: 0.97 | Min Dur: 18s -> Caught: 19 | FA: 11 |
CSI: 0.5588
Window: 70s | Thresh: 0.97 | Min Dur: 19s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 70s | Thresh: 0.97 | Min Dur: 20s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 70s | Thresh: 0.97 | Min Dur: 21s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 70s | Thresh: 0.97 | Min Dur: 22s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 70s | Thresh: 0.97 | Min Dur: 23s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 70s | Thresh: 0.97 | Min Dur: 24s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 70s | Thresh: 0.98 | Min Dur: 5s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 70s | Thresh: 0.98 | Min Dur: 6s -> Caught: 19 | FA: 12 |
CSI: 0.5429
Window: 70s | Thresh: 0.98 | Min Dur: 7s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 70s | Thresh: 0.98 | Min Dur: 8s -> Caught: 19 | FA: 10 |
CSI: 0.5758

```

Window: 70s		Thresh: 0.98		Min Dur: 9s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 70s		Thresh: 0.98		Min Dur: 10s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 70s		Thresh: 0.98		Min Dur: 11s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 70s		Thresh: 0.98		Min Dur: 12s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 70s		Thresh: 0.98		Min Dur: 13s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 70s		Thresh: 0.98		Min Dur: 14s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 70s		Thresh: 0.98		Min Dur: 15s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 70s		Thresh: 0.98		Min Dur: 16s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 70s		Thresh: 0.98		Min Dur: 17s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 70s		Thresh: 0.98		Min Dur: 18s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 70s		Thresh: 0.98		Min Dur: 19s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 70s		Thresh: 0.98		Min Dur: 20s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 70s		Thresh: 0.98		Min Dur: 21s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 70s		Thresh: 0.98		Min Dur: 22s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 70s		Thresh: 0.98		Min Dur: 23s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 70s		Thresh: 0.98		Min Dur: 24s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 70s		Thresh: 0.99		Min Dur: 5s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 70s		Thresh: 0.99		Min Dur: 6s	->	Caught: 15		FA: 8	
CSI: 0.4839									
Window: 70s		Thresh: 0.99		Min Dur: 7s	->	Caught: 15		FA: 8	
CSI: 0.4839									
Window: 70s		Thresh: 0.99		Min Dur: 8s	->	Caught: 15		FA: 8	
CSI: 0.4839									
Window: 70s		Thresh: 0.99		Min Dur: 9s	->	Caught: 15		FA: 9	
CSI: 0.4688									
Window: 70s		Thresh: 0.99		Min Dur: 10s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 70s		Thresh: 0.99		Min Dur: 11s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 70s		Thresh: 0.99		Min Dur: 12s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 70s		Thresh: 0.99		Min Dur: 13s	->	Caught: 15		FA: 7	

CSI: 0.5000					
Window: 70s		Thresh: 0.99		Min Dur: 14s ->	Caught: 15 FA: 7
CSI: 0.5000					
Window: 70s		Thresh: 0.99		Min Dur: 15s ->	Caught: 15 FA: 7
CSI: 0.5000					
Window: 70s		Thresh: 0.99		Min Dur: 16s ->	Caught: 15 FA: 7
CSI: 0.5000					
Window: 70s		Thresh: 0.99		Min Dur: 17s ->	Caught: 15 FA: 7
CSI: 0.5000					
Window: 70s		Thresh: 0.99		Min Dur: 18s ->	Caught: 15 FA: 7
CSI: 0.5000					
Window: 70s		Thresh: 0.99		Min Dur: 19s ->	Caught: 15 FA: 7
CSI: 0.5000					
Window: 70s		Thresh: 0.99		Min Dur: 20s ->	Caught: 15 FA: 7
CSI: 0.5000					
Window: 70s		Thresh: 0.99		Min Dur: 21s ->	Caught: 15 FA: 7
CSI: 0.5000					
Window: 70s		Thresh: 0.99		Min Dur: 22s ->	Caught: 15 FA: 7
CSI: 0.5000					
Window: 70s		Thresh: 0.99		Min Dur: 23s ->	Caught: 15 FA: 7
CSI: 0.5000					
Window: 70s		Thresh: 0.99		Min Dur: 24s ->	Caught: 15 FA: 7
CSI: 0.5000					
Window: 75s		Thresh: 0.97		Min Dur: 5s ->	Caught: 22 FA: 13
CSI: 0.6111					
Window: 75s		Thresh: 0.97		Min Dur: 6s ->	Caught: 22 FA: 12
CSI: 0.6286					
Window: 75s		Thresh: 0.97		Min Dur: 7s ->	Caught: 22 FA: 11
CSI: 0.6471					
Window: 75s		Thresh: 0.97		Min Dur: 8s ->	Caught: 22 FA: 11
CSI: 0.6471					
Window: 75s		Thresh: 0.97		Min Dur: 9s ->	Caught: 21 FA: 11
CSI: 0.6176					
Window: 75s		Thresh: 0.97		Min Dur: 10s ->	Caught: 20 FA: 11
CSI: 0.5882					
Window: 75s		Thresh: 0.97		Min Dur: 11s ->	Caught: 19 FA: 10
CSI: 0.5758					
Window: 75s		Thresh: 0.97		Min Dur: 12s ->	Caught: 19 FA: 10
CSI: 0.5758					
Window: 75s		Thresh: 0.97		Min Dur: 13s ->	Caught: 19 FA: 11
CSI: 0.5588					
Window: 75s		Thresh: 0.97		Min Dur: 14s ->	Caught: 19 FA: 11
CSI: 0.5588					
Window: 75s		Thresh: 0.97		Min Dur: 15s ->	Caught: 19 FA: 10
CSI: 0.5758					
Window: 75s		Thresh: 0.97		Min Dur: 16s ->	Caught: 19 FA: 10
CSI: 0.5758					
Window: 75s		Thresh: 0.97		Min Dur: 17s ->	Caught: 19 FA: 10
CSI: 0.5758					

Window: 75s		Thresh: 0.97		Min Dur: 18s	->	Caught: 19		FA: 11	
CSI: 0.5588									
Window: 75s		Thresh: 0.97		Min Dur: 19s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 75s		Thresh: 0.97		Min Dur: 20s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 75s		Thresh: 0.97		Min Dur: 21s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 75s		Thresh: 0.97		Min Dur: 22s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 75s		Thresh: 0.97		Min Dur: 23s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 75s		Thresh: 0.97		Min Dur: 24s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 75s		Thresh: 0.98		Min Dur: 5s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 75s		Thresh: 0.98		Min Dur: 6s	->	Caught: 19		FA: 11	
CSI: 0.5588									
Window: 75s		Thresh: 0.98		Min Dur: 7s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 75s		Thresh: 0.98		Min Dur: 8s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 75s		Thresh: 0.98		Min Dur: 9s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 75s		Thresh: 0.98		Min Dur: 10s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 75s		Thresh: 0.98		Min Dur: 11s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 75s		Thresh: 0.98		Min Dur: 12s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 75s		Thresh: 0.98		Min Dur: 13s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 75s		Thresh: 0.98		Min Dur: 14s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 75s		Thresh: 0.98		Min Dur: 15s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 75s		Thresh: 0.98		Min Dur: 16s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 75s		Thresh: 0.98		Min Dur: 17s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 75s		Thresh: 0.98		Min Dur: 18s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 75s		Thresh: 0.98		Min Dur: 19s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 75s		Thresh: 0.98		Min Dur: 20s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 75s		Thresh: 0.98		Min Dur: 21s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 75s		Thresh: 0.98		Min Dur: 22s	->	Caught: 19		FA: 8	

```

CSI: 0.6129
Window: 75s | Thresh: 0.98 | Min Dur: 23s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 75s | Thresh: 0.98 | Min Dur: 24s -> Caught: 19 | FA: 7 |
CSI: 0.6333
Window: 75s | Thresh: 0.99 | Min Dur: 5s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 75s | Thresh: 0.99 | Min Dur: 6s -> Caught: 15 | FA: 8 |
CSI: 0.4839
Window: 75s | Thresh: 0.99 | Min Dur: 7s -> Caught: 15 | FA: 8 |
CSI: 0.4839
Window: 75s | Thresh: 0.99 | Min Dur: 8s -> Caught: 15 | FA: 8 |
CSI: 0.4839
Window: 75s | Thresh: 0.99 | Min Dur: 9s -> Caught: 15 | FA: 8 |
CSI: 0.4839
Window: 75s | Thresh: 0.99 | Min Dur: 10s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 75s | Thresh: 0.99 | Min Dur: 11s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 75s | Thresh: 0.99 | Min Dur: 12s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 75s | Thresh: 0.99 | Min Dur: 13s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 75s | Thresh: 0.99 | Min Dur: 14s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 75s | Thresh: 0.99 | Min Dur: 15s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 75s | Thresh: 0.99 | Min Dur: 16s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 75s | Thresh: 0.99 | Min Dur: 17s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 75s | Thresh: 0.99 | Min Dur: 18s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 75s | Thresh: 0.99 | Min Dur: 19s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 75s | Thresh: 0.99 | Min Dur: 20s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 75s | Thresh: 0.99 | Min Dur: 21s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 75s | Thresh: 0.99 | Min Dur: 22s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 75s | Thresh: 0.99 | Min Dur: 23s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 75s | Thresh: 0.99 | Min Dur: 24s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 80s | Thresh: 0.97 | Min Dur: 5s -> Caught: 22 | FA: 13 |
CSI: 0.6111
Window: 80s | Thresh: 0.97 | Min Dur: 6s -> Caught: 22 | FA: 12 |
CSI: 0.6286

```


Window: 80s CSI: 0.6471	Thresh: 0.97	Min Dur: 7s	-> Caught: 22	FA: 11
Window: 80s CSI: 0.6471	Thresh: 0.97	Min Dur: 8s	-> Caught: 22	FA: 11
Window: 80s CSI: 0.6176	Thresh: 0.97	Min Dur: 9s	-> Caught: 21	FA: 11
Window: 80s CSI: 0.5882	Thresh: 0.97	Min Dur: 10s	-> Caught: 20	FA: 11
Window: 80s CSI: 0.5758	Thresh: 0.97	Min Dur: 11s	-> Caught: 19	FA: 10
Window: 80s CSI: 0.5758	Thresh: 0.97	Min Dur: 12s	-> Caught: 19	FA: 10
Window: 80s CSI: 0.5758	Thresh: 0.97	Min Dur: 13s	-> Caught: 19	FA: 10
Window: 80s CSI: 0.5588	Thresh: 0.97	Min Dur: 14s	-> Caught: 19	FA: 11
Window: 80s CSI: 0.5758	Thresh: 0.97	Min Dur: 15s	-> Caught: 19	FA: 10
Window: 80s CSI: 0.5758	Thresh: 0.97	Min Dur: 16s	-> Caught: 19	FA: 10
Window: 80s CSI: 0.5758	Thresh: 0.97	Min Dur: 17s	-> Caught: 19	FA: 10
Window: 80s CSI: 0.5758	Thresh: 0.97	Min Dur: 18s	-> Caught: 19	FA: 10
Window: 80s CSI: 0.5758	Thresh: 0.97	Min Dur: 19s	-> Caught: 19	FA: 10
Window: 80s CSI: 0.5758	Thresh: 0.97	Min Dur: 20s	-> Caught: 19	FA: 10
Window: 80s CSI: 0.5758	Thresh: 0.97	Min Dur: 21s	-> Caught: 19	FA: 10
Window: 80s CSI: 0.5758	Thresh: 0.97	Min Dur: 22s	-> Caught: 19	FA: 10
Window: 80s CSI: 0.5758	Thresh: 0.97	Min Dur: 23s	-> Caught: 19	FA: 10
Window: 80s CSI: 0.5758	Thresh: 0.97	Min Dur: 24s	-> Caught: 19	FA: 10
Window: 80s CSI: 0.5758	Thresh: 0.98	Min Dur: 5s	-> Caught: 19	FA: 10
Window: 80s CSI: 0.5758	Thresh: 0.98	Min Dur: 6s	-> Caught: 19	FA: 10
Window: 80s CSI: 0.5758	Thresh: 0.98	Min Dur: 7s	-> Caught: 19	FA: 10
Window: 80s CSI: 0.5758	Thresh: 0.98	Min Dur: 8s	-> Caught: 19	FA: 10
Window: 80s CSI: 0.5758	Thresh: 0.98	Min Dur: 9s	-> Caught: 19	FA: 10
Window: 80s CSI: 0.5938	Thresh: 0.98	Min Dur: 10s	-> Caught: 19	FA: 9
Window: 80s CSI: 0.5938	Thresh: 0.98	Min Dur: 11s	-> Caught: 19	FA: 9

Window: 80s		Thresh: 0.98		Min Dur: 12s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 80s		Thresh: 0.98		Min Dur: 13s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 80s		Thresh: 0.98		Min Dur: 14s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 80s		Thresh: 0.98		Min Dur: 15s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 80s		Thresh: 0.98		Min Dur: 16s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 80s		Thresh: 0.98		Min Dur: 17s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 80s		Thresh: 0.98		Min Dur: 18s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 80s		Thresh: 0.98		Min Dur: 19s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 80s		Thresh: 0.98		Min Dur: 20s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 80s		Thresh: 0.98		Min Dur: 21s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 80s		Thresh: 0.98		Min Dur: 22s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 80s		Thresh: 0.98		Min Dur: 23s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 80s		Thresh: 0.98		Min Dur: 24s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 80s		Thresh: 0.99		Min Dur: 5s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 80s		Thresh: 0.99		Min Dur: 6s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 80s		Thresh: 0.99		Min Dur: 7s	->	Caught: 15		FA: 8	
CSI: 0.4839									
Window: 80s		Thresh: 0.99		Min Dur: 8s	->	Caught: 15		FA: 8	
CSI: 0.4839									
Window: 80s		Thresh: 0.99		Min Dur: 9s	->	Caught: 15		FA: 8	
CSI: 0.4839									
Window: 80s		Thresh: 0.99		Min Dur: 10s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 80s		Thresh: 0.99		Min Dur: 11s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 80s		Thresh: 0.99		Min Dur: 12s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 80s		Thresh: 0.99		Min Dur: 13s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 80s		Thresh: 0.99		Min Dur: 14s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 80s		Thresh: 0.99		Min Dur: 15s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 80s		Thresh: 0.99		Min Dur: 16s	->	Caught: 15		FA: 7	

CSI: 0.5000				
Window: 80s		Thresh: 0.99		Min Dur: 17s -> Caught: 15 FA: 7
CSI: 0.5000				
Window: 80s		Thresh: 0.99		Min Dur: 18s -> Caught: 15 FA: 7
CSI: 0.5000				
Window: 80s		Thresh: 0.99		Min Dur: 19s -> Caught: 15 FA: 7
CSI: 0.5000				
Window: 80s		Thresh: 0.99		Min Dur: 20s -> Caught: 15 FA: 7
CSI: 0.5000				
Window: 80s		Thresh: 0.99		Min Dur: 21s -> Caught: 15 FA: 7
CSI: 0.5000				
Window: 80s		Thresh: 0.99		Min Dur: 22s -> Caught: 15 FA: 7
CSI: 0.5000				
Window: 80s		Thresh: 0.99		Min Dur: 23s -> Caught: 15 FA: 7
CSI: 0.5000				
Window: 80s		Thresh: 0.99		Min Dur: 24s -> Caught: 15 FA: 7
CSI: 0.5000				
Window: 85s		Thresh: 0.97		Min Dur: 5s -> Caught: 22 FA: 13
CSI: 0.6111				
Window: 85s		Thresh: 0.97		Min Dur: 6s -> Caught: 22 FA: 13
CSI: 0.6111				
Window: 85s		Thresh: 0.97		Min Dur: 7s -> Caught: 22 FA: 11
CSI: 0.6471				
Window: 85s		Thresh: 0.97		Min Dur: 8s -> Caught: 22 FA: 11
CSI: 0.6471				
Window: 85s		Thresh: 0.97		Min Dur: 9s -> Caught: 21 FA: 11
CSI: 0.6176				
Window: 85s		Thresh: 0.97		Min Dur: 10s -> Caught: 20 FA: 11
CSI: 0.5882				
Window: 85s		Thresh: 0.97		Min Dur: 11s -> Caught: 19 FA: 10
CSI: 0.5758				
Window: 85s		Thresh: 0.97		Min Dur: 12s -> Caught: 19 FA: 10
CSI: 0.5758				
Window: 85s		Thresh: 0.97		Min Dur: 13s -> Caught: 19 FA: 10
CSI: 0.5758				
Window: 85s		Thresh: 0.97		Min Dur: 14s -> Caught: 19 FA: 10
CSI: 0.5758				
Window: 85s		Thresh: 0.97		Min Dur: 15s -> Caught: 19 FA: 10
CSI: 0.5758				
Window: 85s		Thresh: 0.97		Min Dur: 16s -> Caught: 19 FA: 10
CSI: 0.5758				
Window: 85s		Thresh: 0.97		Min Dur: 17s -> Caught: 19 FA: 10
CSI: 0.5758				
Window: 85s		Thresh: 0.97		Min Dur: 18s -> Caught: 19 FA: 10
CSI: 0.5758				
Window: 85s		Thresh: 0.97		Min Dur: 19s -> Caught: 19 FA: 9
CSI: 0.5938				
Window: 85s		Thresh: 0.97		Min Dur: 20s -> Caught: 19 FA: 10
CSI: 0.5758				

Window: 85s		Thresh: 0.97		Min Dur: 21s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 85s		Thresh: 0.97		Min Dur: 22s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 85s		Thresh: 0.97		Min Dur: 23s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 85s		Thresh: 0.97		Min Dur: 24s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 85s		Thresh: 0.98		Min Dur: 5s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 85s		Thresh: 0.98		Min Dur: 6s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 85s		Thresh: 0.98		Min Dur: 7s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 85s		Thresh: 0.98		Min Dur: 8s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 85s		Thresh: 0.98		Min Dur: 9s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 85s		Thresh: 0.98		Min Dur: 10s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 85s		Thresh: 0.98		Min Dur: 11s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 85s		Thresh: 0.98		Min Dur: 12s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 85s		Thresh: 0.98		Min Dur: 13s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 85s		Thresh: 0.98		Min Dur: 14s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 85s		Thresh: 0.98		Min Dur: 15s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 85s		Thresh: 0.98		Min Dur: 16s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 85s		Thresh: 0.98		Min Dur: 17s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 85s		Thresh: 0.98		Min Dur: 18s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 85s		Thresh: 0.98		Min Dur: 19s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 85s		Thresh: 0.98		Min Dur: 20s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 85s		Thresh: 0.98		Min Dur: 21s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 85s		Thresh: 0.98		Min Dur: 22s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 85s		Thresh: 0.98		Min Dur: 23s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 85s		Thresh: 0.98		Min Dur: 24s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 85s		Thresh: 0.99		Min Dur: 5s	->	Caught: 15		FA: 7	

```

CSI: 0.5000
Window: 85s | Thresh: 0.99 | Min Dur: 6s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 85s | Thresh: 0.99 | Min Dur: 7s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 85s | Thresh: 0.99 | Min Dur: 8s -> Caught: 15 | FA: 8 |
CSI: 0.4839
Window: 85s | Thresh: 0.99 | Min Dur: 9s -> Caught: 15 | FA: 8 |
CSI: 0.4839
Window: 85s | Thresh: 0.99 | Min Dur: 10s -> Caught: 15 | FA: 8 |
CSI: 0.4839
Window: 85s | Thresh: 0.99 | Min Dur: 11s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 85s | Thresh: 0.99 | Min Dur: 12s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 85s | Thresh: 0.99 | Min Dur: 13s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 85s | Thresh: 0.99 | Min Dur: 14s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 85s | Thresh: 0.99 | Min Dur: 15s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 85s | Thresh: 0.99 | Min Dur: 16s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 85s | Thresh: 0.99 | Min Dur: 17s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 85s | Thresh: 0.99 | Min Dur: 18s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 85s | Thresh: 0.99 | Min Dur: 19s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 85s | Thresh: 0.99 | Min Dur: 20s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 85s | Thresh: 0.99 | Min Dur: 21s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 85s | Thresh: 0.99 | Min Dur: 22s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 85s | Thresh: 0.99 | Min Dur: 23s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 85s | Thresh: 0.99 | Min Dur: 24s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 90s | Thresh: 0.97 | Min Dur: 5s -> Caught: 22 | FA: 13 |
CSI: 0.6111
Window: 90s | Thresh: 0.97 | Min Dur: 6s -> Caught: 22 | FA: 13 |
CSI: 0.6111
Window: 90s | Thresh: 0.97 | Min Dur: 7s -> Caught: 22 | FA: 11 |
CSI: 0.6471
Window: 90s | Thresh: 0.97 | Min Dur: 8s -> Caught: 22 | FA: 11 |
CSI: 0.6471
Window: 90s | Thresh: 0.97 | Min Dur: 9s -> Caught: 21 | FA: 11 |
CSI: 0.6176

```

Window: 90s		Thresh: 0.97		Min Dur: 10s	->	Caught: 20		FA: 11	
CSI: 0.5882									
Window: 90s		Thresh: 0.97		Min Dur: 11s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 90s		Thresh: 0.97		Min Dur: 12s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 90s		Thresh: 0.97		Min Dur: 13s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 90s		Thresh: 0.97		Min Dur: 14s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 90s		Thresh: 0.97		Min Dur: 15s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 90s		Thresh: 0.97		Min Dur: 16s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 90s		Thresh: 0.97		Min Dur: 17s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 90s		Thresh: 0.97		Min Dur: 18s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 90s		Thresh: 0.97		Min Dur: 19s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 90s		Thresh: 0.97		Min Dur: 20s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 90s		Thresh: 0.97		Min Dur: 21s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 90s		Thresh: 0.97		Min Dur: 22s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 90s		Thresh: 0.97		Min Dur: 23s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 90s		Thresh: 0.97		Min Dur: 24s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 90s		Thresh: 0.98		Min Dur: 5s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 90s		Thresh: 0.98		Min Dur: 6s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 90s		Thresh: 0.98		Min Dur: 7s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 90s		Thresh: 0.98		Min Dur: 8s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 90s		Thresh: 0.98		Min Dur: 9s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 90s		Thresh: 0.98		Min Dur: 10s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 90s		Thresh: 0.98		Min Dur: 11s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 90s		Thresh: 0.98		Min Dur: 12s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 90s		Thresh: 0.98		Min Dur: 13s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 90s		Thresh: 0.98		Min Dur: 14s	->	Caught: 19		FA: 8	

CSI: 0.6129	
Window: 90s	Thresh: 0.98 Min Dur: 15s -> Caught: 19 FA: 8
CSI: 0.6129	
Window: 90s	Thresh: 0.98 Min Dur: 16s -> Caught: 19 FA: 8
CSI: 0.6129	
Window: 90s	Thresh: 0.98 Min Dur: 17s -> Caught: 19 FA: 8
CSI: 0.6129	
Window: 90s	Thresh: 0.98 Min Dur: 18s -> Caught: 19 FA: 8
CSI: 0.6129	
Window: 90s	Thresh: 0.98 Min Dur: 19s -> Caught: 19 FA: 8
CSI: 0.6129	
Window: 90s	Thresh: 0.98 Min Dur: 20s -> Caught: 19 FA: 8
CSI: 0.6129	
Window: 90s	Thresh: 0.98 Min Dur: 21s -> Caught: 19 FA: 8
CSI: 0.6129	
Window: 90s	Thresh: 0.98 Min Dur: 22s -> Caught: 19 FA: 8
CSI: 0.6129	
Window: 90s	Thresh: 0.98 Min Dur: 23s -> Caught: 19 FA: 8
CSI: 0.6129	
Window: 90s	Thresh: 0.98 Min Dur: 24s -> Caught: 19 FA: 8
CSI: 0.6129	
Window: 90s	Thresh: 0.99 Min Dur: 5s -> Caught: 15 FA: 7
CSI: 0.5000	
Window: 90s	Thresh: 0.99 Min Dur: 6s -> Caught: 15 FA: 7
CSI: 0.5000	
Window: 90s	Thresh: 0.99 Min Dur: 7s -> Caught: 15 FA: 7
CSI: 0.5000	
Window: 90s	Thresh: 0.99 Min Dur: 8s -> Caught: 15 FA: 8
CSI: 0.4839	
Window: 90s	Thresh: 0.99 Min Dur: 9s -> Caught: 15 FA: 8
CSI: 0.4839	
Window: 90s	Thresh: 0.99 Min Dur: 10s -> Caught: 15 FA: 8
CSI: 0.4839	
Window: 90s	Thresh: 0.99 Min Dur: 11s -> Caught: 15 FA: 7
CSI: 0.5000	
Window: 90s	Thresh: 0.99 Min Dur: 12s -> Caught: 15 FA: 7
CSI: 0.5000	
Window: 90s	Thresh: 0.99 Min Dur: 13s -> Caught: 15 FA: 7
CSI: 0.5000	
Window: 90s	Thresh: 0.99 Min Dur: 14s -> Caught: 15 FA: 7
CSI: 0.5000	
Window: 90s	Thresh: 0.99 Min Dur: 15s -> Caught: 15 FA: 7
CSI: 0.5000	
Window: 90s	Thresh: 0.99 Min Dur: 16s -> Caught: 15 FA: 7
CSI: 0.5000	
Window: 90s	Thresh: 0.99 Min Dur: 17s -> Caught: 15 FA: 7
CSI: 0.5000	
Window: 90s	Thresh: 0.99 Min Dur: 18s -> Caught: 15 FA: 7
CSI: 0.5000	

Window: 90s		Thresh: 0.99		Min Dur: 19s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 90s		Thresh: 0.99		Min Dur: 20s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 90s		Thresh: 0.99		Min Dur: 21s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 90s		Thresh: 0.99		Min Dur: 22s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 90s		Thresh: 0.99		Min Dur: 23s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 90s		Thresh: 0.99		Min Dur: 24s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 95s		Thresh: 0.97		Min Dur: 5s	->	Caught: 22		FA: 12	
CSI: 0.6286									
Window: 95s		Thresh: 0.97		Min Dur: 6s	->	Caught: 22		FA: 13	
CSI: 0.6111									
Window: 95s		Thresh: 0.97		Min Dur: 7s	->	Caught: 22		FA: 11	
CSI: 0.6471									
Window: 95s		Thresh: 0.97		Min Dur: 8s	->	Caught: 22		FA: 11	
CSI: 0.6471									
Window: 95s		Thresh: 0.97		Min Dur: 9s	->	Caught: 22		FA: 11	
CSI: 0.6471									
Window: 95s		Thresh: 0.97		Min Dur: 10s	->	Caught: 21		FA: 11	
CSI: 0.6176									
Window: 95s		Thresh: 0.97		Min Dur: 11s	->	Caught: 19		FA: 11	
CSI: 0.5588									
Window: 95s		Thresh: 0.97		Min Dur: 12s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 95s		Thresh: 0.97		Min Dur: 13s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 95s		Thresh: 0.97		Min Dur: 14s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 95s		Thresh: 0.97		Min Dur: 15s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 95s		Thresh: 0.97		Min Dur: 16s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 95s		Thresh: 0.97		Min Dur: 17s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 95s		Thresh: 0.97		Min Dur: 18s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 95s		Thresh: 0.97		Min Dur: 19s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 95s		Thresh: 0.97		Min Dur: 20s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 95s		Thresh: 0.97		Min Dur: 21s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 95s		Thresh: 0.97		Min Dur: 22s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 95s		Thresh: 0.97		Min Dur: 23s	->	Caught: 19		FA: 10	

CSI: 0.5758				
Window: 95s		Thresh: 0.97		Min Dur: 24s -> Caught: 19 FA: 10
CSI: 0.5758				
Window: 95s		Thresh: 0.98		Min Dur: 5s -> Caught: 19 FA: 9
CSI: 0.5938				
Window: 95s		Thresh: 0.98		Min Dur: 6s -> Caught: 19 FA: 10
CSI: 0.5758				
Window: 95s		Thresh: 0.98		Min Dur: 7s -> Caught: 19 FA: 9
CSI: 0.5938				
Window: 95s		Thresh: 0.98		Min Dur: 8s -> Caught: 19 FA: 11
CSI: 0.5588				
Window: 95s		Thresh: 0.98		Min Dur: 9s -> Caught: 19 FA: 10
CSI: 0.5758				
Window: 95s		Thresh: 0.98		Min Dur: 10s -> Caught: 19 FA: 10
CSI: 0.5758				
Window: 95s		Thresh: 0.98		Min Dur: 11s -> Caught: 19 FA: 9
CSI: 0.5938				
Window: 95s		Thresh: 0.98		Min Dur: 12s -> Caught: 19 FA: 9
CSI: 0.5938				
Window: 95s		Thresh: 0.98		Min Dur: 13s -> Caught: 19 FA: 8
CSI: 0.6129				
Window: 95s		Thresh: 0.98		Min Dur: 14s -> Caught: 19 FA: 8
CSI: 0.6129				
Window: 95s		Thresh: 0.98		Min Dur: 15s -> Caught: 19 FA: 8
CSI: 0.6129				
Window: 95s		Thresh: 0.98		Min Dur: 16s -> Caught: 19 FA: 8
CSI: 0.6129				
Window: 95s		Thresh: 0.98		Min Dur: 17s -> Caught: 19 FA: 8
CSI: 0.6129				
Window: 95s		Thresh: 0.98		Min Dur: 18s -> Caught: 19 FA: 8
CSI: 0.6129				
Window: 95s		Thresh: 0.98		Min Dur: 19s -> Caught: 19 FA: 8
CSI: 0.6129				
Window: 95s		Thresh: 0.98		Min Dur: 20s -> Caught: 19 FA: 8
CSI: 0.6129				
Window: 95s		Thresh: 0.98		Min Dur: 21s -> Caught: 19 FA: 8
CSI: 0.6129				
Window: 95s		Thresh: 0.98		Min Dur: 22s -> Caught: 19 FA: 8
CSI: 0.6129				
Window: 95s		Thresh: 0.98		Min Dur: 23s -> Caught: 19 FA: 8
CSI: 0.6129				
Window: 95s		Thresh: 0.98		Min Dur: 24s -> Caught: 19 FA: 8
CSI: 0.6129				
Window: 95s		Thresh: 0.99		Min Dur: 5s -> Caught: 15 FA: 7
CSI: 0.5000				
Window: 95s		Thresh: 0.99		Min Dur: 6s -> Caught: 15 FA: 7
CSI: 0.5000				
Window: 95s		Thresh: 0.99		Min Dur: 7s -> Caught: 15 FA: 7
CSI: 0.5000				

Window: 95s	Thresh: 0.99	Min Dur: 8s	-> Caught: 15	FA: 8
CSI: 0.4839				
Window: 95s	Thresh: 0.99	Min Dur: 9s	-> Caught: 15	FA: 8
CSI: 0.4839				
Window: 95s	Thresh: 0.99	Min Dur: 10s	-> Caught: 15	FA: 8
CSI: 0.4839				
Window: 95s	Thresh: 0.99	Min Dur: 11s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 95s	Thresh: 0.99	Min Dur: 12s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 95s	Thresh: 0.99	Min Dur: 13s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 95s	Thresh: 0.99	Min Dur: 14s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 95s	Thresh: 0.99	Min Dur: 15s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 95s	Thresh: 0.99	Min Dur: 16s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 95s	Thresh: 0.99	Min Dur: 17s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 95s	Thresh: 0.99	Min Dur: 18s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 95s	Thresh: 0.99	Min Dur: 19s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 95s	Thresh: 0.99	Min Dur: 20s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 95s	Thresh: 0.99	Min Dur: 21s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 95s	Thresh: 0.99	Min Dur: 22s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 95s	Thresh: 0.99	Min Dur: 23s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 95s	Thresh: 0.99	Min Dur: 24s	-> Caught: 15	FA: 7
CSI: 0.5000				
Window: 100s	Thresh: 0.97	Min Dur: 5s	-> Caught: 22	FA: 12
CSI: 0.6286				
Window: 100s	Thresh: 0.97	Min Dur: 6s	-> Caught: 22	FA: 12
CSI: 0.6286				
Window: 100s	Thresh: 0.97	Min Dur: 7s	-> Caught: 22	FA: 11
CSI: 0.6471				
Window: 100s	Thresh: 0.97	Min Dur: 8s	-> Caught: 22	FA: 11
CSI: 0.6471				
Window: 100s	Thresh: 0.97	Min Dur: 9s	-> Caught: 22	FA: 11
CSI: 0.6471				
Window: 100s	Thresh: 0.97	Min Dur: 10s	-> Caught: 21	FA: 11
CSI: 0.6176				
Window: 100s	Thresh: 0.97	Min Dur: 11s	-> Caught: 21	FA: 11
CSI: 0.6176				
Window: 100s	Thresh: 0.97	Min Dur: 12s	-> Caught: 20	FA: 10

CSI: 0.6061
Window: 100s | Thresh: 0.97 | Min Dur: 13s -> Caught: 20 | FA: 10 |
CSI: 0.6061
Window: 100s | Thresh: 0.97 | Min Dur: 14s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 100s | Thresh: 0.97 | Min Dur: 15s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 100s | Thresh: 0.97 | Min Dur: 16s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 100s | Thresh: 0.97 | Min Dur: 17s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 100s | Thresh: 0.97 | Min Dur: 18s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 100s | Thresh: 0.97 | Min Dur: 19s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 100s | Thresh: 0.97 | Min Dur: 20s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 100s | Thresh: 0.97 | Min Dur: 21s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 100s | Thresh: 0.97 | Min Dur: 22s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 100s | Thresh: 0.97 | Min Dur: 23s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 100s | Thresh: 0.97 | Min Dur: 24s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 100s | Thresh: 0.98 | Min Dur: 5s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 100s | Thresh: 0.98 | Min Dur: 6s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 100s | Thresh: 0.98 | Min Dur: 7s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 100s | Thresh: 0.98 | Min Dur: 8s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 100s | Thresh: 0.98 | Min Dur: 9s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 100s | Thresh: 0.98 | Min Dur: 10s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 100s | Thresh: 0.98 | Min Dur: 11s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 100s | Thresh: 0.98 | Min Dur: 12s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 100s | Thresh: 0.98 | Min Dur: 13s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 100s | Thresh: 0.98 | Min Dur: 14s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 100s | Thresh: 0.98 | Min Dur: 15s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 100s | Thresh: 0.98 | Min Dur: 16s -> Caught: 19 | FA: 8 |
CSI: 0.6129

Window: 100s		Thresh: 0.98		Min Dur: 17s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 100s		Thresh: 0.98		Min Dur: 18s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 100s		Thresh: 0.98		Min Dur: 19s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 100s		Thresh: 0.98		Min Dur: 20s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 100s		Thresh: 0.98		Min Dur: 21s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 100s		Thresh: 0.98		Min Dur: 22s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 100s		Thresh: 0.98		Min Dur: 23s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 100s		Thresh: 0.98		Min Dur: 24s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 100s		Thresh: 0.99		Min Dur: 5s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 100s		Thresh: 0.99		Min Dur: 6s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 100s		Thresh: 0.99		Min Dur: 7s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 100s		Thresh: 0.99		Min Dur: 8s	->	Caught: 15		FA: 8	
CSI: 0.4839									
Window: 100s		Thresh: 0.99		Min Dur: 9s	->	Caught: 15		FA: 8	
CSI: 0.4839									
Window: 100s		Thresh: 0.99		Min Dur: 10s	->	Caught: 15		FA: 8	
CSI: 0.4839									
Window: 100s		Thresh: 0.99		Min Dur: 11s	->	Caught: 15		FA: 8	
CSI: 0.4839									
Window: 100s		Thresh: 0.99		Min Dur: 12s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 100s		Thresh: 0.99		Min Dur: 13s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 100s		Thresh: 0.99		Min Dur: 14s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 100s		Thresh: 0.99		Min Dur: 15s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 100s		Thresh: 0.99		Min Dur: 16s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 100s		Thresh: 0.99		Min Dur: 17s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 100s		Thresh: 0.99		Min Dur: 18s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 100s		Thresh: 0.99		Min Dur: 19s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 100s		Thresh: 0.99		Min Dur: 20s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 100s		Thresh: 0.99		Min Dur: 21s	->	Caught: 15		FA: 7	
CSI: 0.5000									

Window: 100s		Thresh: 0.99		Min Dur: 22s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 100s		Thresh: 0.99		Min Dur: 23s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 100s		Thresh: 0.99		Min Dur: 24s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 105s		Thresh: 0.97		Min Dur: 5s	->	Caught: 22		FA: 12	
CSI: 0.6286									
Window: 105s		Thresh: 0.97		Min Dur: 6s	->	Caught: 22		FA: 12	
CSI: 0.6286									
Window: 105s		Thresh: 0.97		Min Dur: 7s	->	Caught: 22		FA: 11	
CSI: 0.6471									
Window: 105s		Thresh: 0.97		Min Dur: 8s	->	Caught: 22		FA: 11	
CSI: 0.6471									
Window: 105s		Thresh: 0.97		Min Dur: 9s	->	Caught: 22		FA: 11	
CSI: 0.6471									
Window: 105s		Thresh: 0.97		Min Dur: 10s	->	Caught: 21		FA: 11	
CSI: 0.6176									
Window: 105s		Thresh: 0.97		Min Dur: 11s	->	Caught: 21		FA: 11	
CSI: 0.6176									
Window: 105s		Thresh: 0.97		Min Dur: 12s	->	Caught: 20		FA: 11	
CSI: 0.5882									
Window: 105s		Thresh: 0.97		Min Dur: 13s	->	Caught: 20		FA: 10	
CSI: 0.6061									
Window: 105s		Thresh: 0.97		Min Dur: 14s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 105s		Thresh: 0.97		Min Dur: 15s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 105s		Thresh: 0.97		Min Dur: 16s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 105s		Thresh: 0.97		Min Dur: 17s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 105s		Thresh: 0.97		Min Dur: 18s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 105s		Thresh: 0.97		Min Dur: 19s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 105s		Thresh: 0.97		Min Dur: 20s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 105s		Thresh: 0.97		Min Dur: 21s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 105s		Thresh: 0.97		Min Dur: 22s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 105s		Thresh: 0.97		Min Dur: 23s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 105s		Thresh: 0.97		Min Dur: 24s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 105s		Thresh: 0.98		Min Dur: 5s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 105s		Thresh: 0.98		Min Dur: 6s	->	Caught: 19		FA: 10	

CSI: 0.5758
Window: 105s | Thresh: 0.98 | Min Dur: 7s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 105s | Thresh: 0.98 | Min Dur: 8s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 105s | Thresh: 0.98 | Min Dur: 9s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 105s | Thresh: 0.98 | Min Dur: 10s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 105s | Thresh: 0.98 | Min Dur: 11s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 105s | Thresh: 0.98 | Min Dur: 12s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 105s | Thresh: 0.98 | Min Dur: 13s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 105s | Thresh: 0.98 | Min Dur: 14s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 105s | Thresh: 0.98 | Min Dur: 15s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 105s | Thresh: 0.98 | Min Dur: 16s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 105s | Thresh: 0.98 | Min Dur: 17s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 105s | Thresh: 0.98 | Min Dur: 18s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 105s | Thresh: 0.98 | Min Dur: 19s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 105s | Thresh: 0.98 | Min Dur: 20s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 105s | Thresh: 0.98 | Min Dur: 21s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 105s | Thresh: 0.98 | Min Dur: 22s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 105s | Thresh: 0.98 | Min Dur: 23s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 105s | Thresh: 0.98 | Min Dur: 24s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 105s | Thresh: 0.99 | Min Dur: 5s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 105s | Thresh: 0.99 | Min Dur: 6s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 105s | Thresh: 0.99 | Min Dur: 7s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 105s | Thresh: 0.99 | Min Dur: 8s -> Caught: 15 | FA: 8 |
CSI: 0.4839
Window: 105s | Thresh: 0.99 | Min Dur: 9s -> Caught: 15 | FA: 8 |
CSI: 0.4839
Window: 105s | Thresh: 0.99 | Min Dur: 10s -> Caught: 15 | FA: 8 |
CSI: 0.4839

Window: 105s		Thresh: 0.99		Min Dur: 11s	->	Caught: 15		FA: 8	
CSI: 0.4839									
Window: 105s		Thresh: 0.99		Min Dur: 12s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 105s		Thresh: 0.99		Min Dur: 13s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 105s		Thresh: 0.99		Min Dur: 14s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 105s		Thresh: 0.99		Min Dur: 15s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 105s		Thresh: 0.99		Min Dur: 16s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 105s		Thresh: 0.99		Min Dur: 17s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 105s		Thresh: 0.99		Min Dur: 18s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 105s		Thresh: 0.99		Min Dur: 19s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 105s		Thresh: 0.99		Min Dur: 20s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 105s		Thresh: 0.99		Min Dur: 21s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 105s		Thresh: 0.99		Min Dur: 22s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 105s		Thresh: 0.99		Min Dur: 23s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 105s		Thresh: 0.99		Min Dur: 24s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 110s		Thresh: 0.97		Min Dur: 5s	->	Caught: 22		FA: 12	
CSI: 0.6286									
Window: 110s		Thresh: 0.97		Min Dur: 6s	->	Caught: 22		FA: 12	
CSI: 0.6286									
Window: 110s		Thresh: 0.97		Min Dur: 7s	->	Caught: 22		FA: 10	
CSI: 0.6667									
Window: 110s		Thresh: 0.97		Min Dur: 8s	->	Caught: 22		FA: 11	
CSI: 0.6471									
Window: 110s		Thresh: 0.97		Min Dur: 9s	->	Caught: 22		FA: 11	
CSI: 0.6471									
Window: 110s		Thresh: 0.97		Min Dur: 10s	->	Caught: 21		FA: 11	
CSI: 0.6176									
Window: 110s		Thresh: 0.97		Min Dur: 11s	->	Caught: 21		FA: 11	
CSI: 0.6176									
Window: 110s		Thresh: 0.97		Min Dur: 12s	->	Caught: 20		FA: 11	
CSI: 0.5882									
Window: 110s		Thresh: 0.97		Min Dur: 13s	->	Caught: 20		FA: 10	
CSI: 0.6061									
Window: 110s		Thresh: 0.97		Min Dur: 14s	->	Caught: 19		FA: 10	
CSI: 0.5758									
Window: 110s		Thresh: 0.97		Min Dur: 15s	->	Caught: 19		FA: 10	

CSI: 0.5758
 Window: 110s | Thresh: 0.97 | Min Dur: 16s -> Caught: 19 | FA: 9 |
 CSI: 0.5938
 Window: 110s | Thresh: 0.97 | Min Dur: 17s -> Caught: 19 | FA: 9 |
 CSI: 0.5938
 Window: 110s | Thresh: 0.97 | Min Dur: 18s -> Caught: 19 | FA: 9 |
 CSI: 0.5938
 Window: 110s | Thresh: 0.97 | Min Dur: 19s -> Caught: 19 | FA: 8 |
 CSI: 0.6129
 Window: 110s | Thresh: 0.97 | Min Dur: 20s -> Caught: 19 | FA: 9 |
 CSI: 0.5938
 Window: 110s | Thresh: 0.97 | Min Dur: 21s -> Caught: 19 | FA: 9 |
 CSI: 0.5938
 Window: 110s | Thresh: 0.97 | Min Dur: 22s -> Caught: 19 | FA: 9 |
 CSI: 0.5938
 Window: 110s | Thresh: 0.97 | Min Dur: 23s -> Caught: 19 | FA: 9 |
 CSI: 0.5938
 Window: 110s | Thresh: 0.97 | Min Dur: 24s -> Caught: 19 | FA: 9 |
 CSI: 0.5938
 Window: 110s | Thresh: 0.98 | Min Dur: 5s -> Caught: 19 | FA: 8 |
 CSI: 0.6129
 Window: 110s | Thresh: 0.98 | Min Dur: 6s -> Caught: 19 | FA: 10 |
 CSI: 0.5758
 Window: 110s | Thresh: 0.98 | Min Dur: 7s -> Caught: 19 | FA: 9 |
 CSI: 0.5938
 Window: 110s | Thresh: 0.98 | Min Dur: 8s -> Caught: 19 | FA: 9 |
 CSI: 0.5938
 Window: 110s | Thresh: 0.98 | Min Dur: 9s -> Caught: 19 | FA: 9 |
 CSI: 0.5938
 Window: 110s | Thresh: 0.98 | Min Dur: 10s -> Caught: 19 | FA: 9 |
 CSI: 0.5938
 Window: 110s | Thresh: 0.98 | Min Dur: 11s -> Caught: 19 | FA: 9 |
 CSI: 0.5938
 Window: 110s | Thresh: 0.98 | Min Dur: 12s -> Caught: 19 | FA: 10 |
 CSI: 0.5758
 Window: 110s | Thresh: 0.98 | Min Dur: 13s -> Caught: 19 | FA: 9 |
 CSI: 0.5938
 Window: 110s | Thresh: 0.98 | Min Dur: 14s -> Caught: 19 | FA: 8 |
 CSI: 0.6129
 Window: 110s | Thresh: 0.98 | Min Dur: 15s -> Caught: 19 | FA: 8 |
 CSI: 0.6129
 Window: 110s | Thresh: 0.98 | Min Dur: 16s -> Caught: 19 | FA: 8 |
 CSI: 0.6129
 Window: 110s | Thresh: 0.98 | Min Dur: 17s -> Caught: 19 | FA: 8 |
 CSI: 0.6129
 Window: 110s | Thresh: 0.98 | Min Dur: 18s -> Caught: 19 | FA: 8 |
 CSI: 0.6129
 Window: 110s | Thresh: 0.98 | Min Dur: 19s -> Caught: 19 | FA: 8 |
 CSI: 0.6129

Window: 110s		Thresh: 0.98		Min Dur: 20s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 110s		Thresh: 0.98		Min Dur: 21s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 110s		Thresh: 0.98		Min Dur: 22s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 110s		Thresh: 0.98		Min Dur: 23s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 110s		Thresh: 0.98		Min Dur: 24s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 110s		Thresh: 0.99		Min Dur: 5s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 110s		Thresh: 0.99		Min Dur: 6s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 110s		Thresh: 0.99		Min Dur: 7s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 110s		Thresh: 0.99		Min Dur: 8s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 110s		Thresh: 0.99		Min Dur: 9s	->	Caught: 15		FA: 8	
CSI: 0.4839									
Window: 110s		Thresh: 0.99		Min Dur: 10s	->	Caught: 15		FA: 8	
CSI: 0.4839									
Window: 110s		Thresh: 0.99		Min Dur: 11s	->	Caught: 15		FA: 8	
CSI: 0.4839									
Window: 110s		Thresh: 0.99		Min Dur: 12s	->	Caught: 15		FA: 8	
CSI: 0.4839									
Window: 110s		Thresh: 0.99		Min Dur: 13s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 110s		Thresh: 0.99		Min Dur: 14s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 110s		Thresh: 0.99		Min Dur: 15s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 110s		Thresh: 0.99		Min Dur: 16s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 110s		Thresh: 0.99		Min Dur: 17s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 110s		Thresh: 0.99		Min Dur: 18s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 110s		Thresh: 0.99		Min Dur: 19s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 110s		Thresh: 0.99		Min Dur: 20s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 110s		Thresh: 0.99		Min Dur: 21s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 110s		Thresh: 0.99		Min Dur: 22s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 110s		Thresh: 0.99		Min Dur: 23s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 110s		Thresh: 0.99		Min Dur: 24s	->	Caught: 15		FA: 7	

CSI: 0.5000
Window: 115s | Thresh: 0.97 | Min Dur: 5s -> Caught: 22 | FA: 13 |
CSI: 0.6111
Window: 115s | Thresh: 0.97 | Min Dur: 6s -> Caught: 22 | FA: 12 |
CSI: 0.6286
Window: 115s | Thresh: 0.97 | Min Dur: 7s -> Caught: 22 | FA: 10 |
CSI: 0.6667
Window: 115s | Thresh: 0.97 | Min Dur: 8s -> Caught: 22 | FA: 11 |
CSI: 0.6471
Window: 115s | Thresh: 0.97 | Min Dur: 9s -> Caught: 22 | FA: 11 |
CSI: 0.6471
Window: 115s | Thresh: 0.97 | Min Dur: 10s -> Caught: 21 | FA: 11 |
CSI: 0.6176
Window: 115s | Thresh: 0.97 | Min Dur: 11s -> Caught: 21 | FA: 11 |
CSI: 0.6176
Window: 115s | Thresh: 0.97 | Min Dur: 12s -> Caught: 20 | FA: 11 |
CSI: 0.5882
Window: 115s | Thresh: 0.97 | Min Dur: 13s -> Caught: 20 | FA: 10 |
CSI: 0.6061
Window: 115s | Thresh: 0.97 | Min Dur: 14s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 115s | Thresh: 0.97 | Min Dur: 15s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 115s | Thresh: 0.97 | Min Dur: 16s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 115s | Thresh: 0.97 | Min Dur: 17s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 115s | Thresh: 0.97 | Min Dur: 18s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 115s | Thresh: 0.97 | Min Dur: 19s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 115s | Thresh: 0.97 | Min Dur: 20s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 115s | Thresh: 0.97 | Min Dur: 21s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 115s | Thresh: 0.97 | Min Dur: 22s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 115s | Thresh: 0.97 | Min Dur: 23s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 115s | Thresh: 0.97 | Min Dur: 24s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 115s | Thresh: 0.98 | Min Dur: 5s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 115s | Thresh: 0.98 | Min Dur: 6s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 115s | Thresh: 0.98 | Min Dur: 7s -> Caught: 19 | FA: 9 |
CSI: 0.5938
Window: 115s | Thresh: 0.98 | Min Dur: 8s -> Caught: 19 | FA: 9 |
CSI: 0.5938

Window: 115s		Thresh: 0.98		Min Dur: 9s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 115s		Thresh: 0.98		Min Dur: 10s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 115s		Thresh: 0.98		Min Dur: 11s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 115s		Thresh: 0.98		Min Dur: 12s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 115s		Thresh: 0.98		Min Dur: 13s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 115s		Thresh: 0.98		Min Dur: 14s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 115s		Thresh: 0.98		Min Dur: 15s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 115s		Thresh: 0.98		Min Dur: 16s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 115s		Thresh: 0.98		Min Dur: 17s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 115s		Thresh: 0.98		Min Dur: 18s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 115s		Thresh: 0.98		Min Dur: 19s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 115s		Thresh: 0.98		Min Dur: 20s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 115s		Thresh: 0.98		Min Dur: 21s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 115s		Thresh: 0.98		Min Dur: 22s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 115s		Thresh: 0.98		Min Dur: 23s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 115s		Thresh: 0.98		Min Dur: 24s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 115s		Thresh: 0.99		Min Dur: 5s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 115s		Thresh: 0.99		Min Dur: 6s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 115s		Thresh: 0.99		Min Dur: 7s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 115s		Thresh: 0.99		Min Dur: 8s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 115s		Thresh: 0.99		Min Dur: 9s	->	Caught: 15		FA: 7	
CSI: 0.5000									
Window: 115s		Thresh: 0.99		Min Dur: 10s	->	Caught: 15		FA: 8	
CSI: 0.4839									
Window: 115s		Thresh: 0.99		Min Dur: 11s	->	Caught: 15		FA: 8	
CSI: 0.4839									
Window: 115s		Thresh: 0.99		Min Dur: 12s	->	Caught: 15		FA: 9	
CSI: 0.4688									
Window: 115s		Thresh: 0.99		Min Dur: 13s	->	Caught: 15		FA: 7	

```

CSI: 0.5000
Window: 115s | Thresh: 0.99 | Min Dur: 14s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 115s | Thresh: 0.99 | Min Dur: 15s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 115s | Thresh: 0.99 | Min Dur: 16s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 115s | Thresh: 0.99 | Min Dur: 17s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 115s | Thresh: 0.99 | Min Dur: 18s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 115s | Thresh: 0.99 | Min Dur: 19s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 115s | Thresh: 0.99 | Min Dur: 20s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 115s | Thresh: 0.99 | Min Dur: 21s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 115s | Thresh: 0.99 | Min Dur: 22s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 115s | Thresh: 0.99 | Min Dur: 23s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 115s | Thresh: 0.99 | Min Dur: 24s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 120s | Thresh: 0.97 | Min Dur: 5s -> Caught: 22 | FA: 12 |
CSI: 0.6286
Window: 120s | Thresh: 0.97 | Min Dur: 6s -> Caught: 22 | FA: 12 |
CSI: 0.6286
Window: 120s | Thresh: 0.97 | Min Dur: 7s -> Caught: 22 | FA: 11 |
CSI: 0.6471
Window: 120s | Thresh: 0.97 | Min Dur: 8s -> Caught: 22 | FA: 11 |
CSI: 0.6471
Window: 120s | Thresh: 0.97 | Min Dur: 9s -> Caught: 22 | FA: 11 |
CSI: 0.6471
Window: 120s | Thresh: 0.97 | Min Dur: 10s -> Caught: 21 | FA: 11 |
CSI: 0.6176
Window: 120s | Thresh: 0.97 | Min Dur: 11s -> Caught: 21 | FA: 11 |
CSI: 0.6176
Window: 120s | Thresh: 0.97 | Min Dur: 12s -> Caught: 20 | FA: 11 |
CSI: 0.5882
Window: 120s | Thresh: 0.97 | Min Dur: 13s -> Caught: 20 | FA: 10 |
CSI: 0.6061
Window: 120s | Thresh: 0.97 | Min Dur: 14s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 120s | Thresh: 0.97 | Min Dur: 15s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 120s | Thresh: 0.97 | Min Dur: 16s -> Caught: 19 | FA: 10 |
CSI: 0.5758
Window: 120s | Thresh: 0.97 | Min Dur: 17s -> Caught: 19 | FA: 9 |
CSI: 0.5938

```

Window: 120s		Thresh: 0.97		Min Dur: 18s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 120s		Thresh: 0.97		Min Dur: 19s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 120s		Thresh: 0.97		Min Dur: 20s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 120s		Thresh: 0.97		Min Dur: 21s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 120s		Thresh: 0.97		Min Dur: 22s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 120s		Thresh: 0.97		Min Dur: 23s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 120s		Thresh: 0.97		Min Dur: 24s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 120s		Thresh: 0.98		Min Dur: 5s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 120s		Thresh: 0.98		Min Dur: 6s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 120s		Thresh: 0.98		Min Dur: 7s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 120s		Thresh: 0.98		Min Dur: 8s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 120s		Thresh: 0.98		Min Dur: 9s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 120s		Thresh: 0.98		Min Dur: 10s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 120s		Thresh: 0.98		Min Dur: 11s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 120s		Thresh: 0.98		Min Dur: 12s	->	Caught: 19		FA: 9	
CSI: 0.5938									
Window: 120s		Thresh: 0.98		Min Dur: 13s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 120s		Thresh: 0.98		Min Dur: 14s	->	Caught: 19		FA: 7	
CSI: 0.6333									
Window: 120s		Thresh: 0.98		Min Dur: 15s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 120s		Thresh: 0.98		Min Dur: 16s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 120s		Thresh: 0.98		Min Dur: 17s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 120s		Thresh: 0.98		Min Dur: 18s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 120s		Thresh: 0.98		Min Dur: 19s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 120s		Thresh: 0.98		Min Dur: 20s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 120s		Thresh: 0.98		Min Dur: 21s	->	Caught: 19		FA: 8	
CSI: 0.6129									
Window: 120s		Thresh: 0.98		Min Dur: 22s	->	Caught: 19		FA: 8	

```

CSI: 0.6129
Window: 120s | Thresh: 0.98 | Min Dur: 23s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 120s | Thresh: 0.98 | Min Dur: 24s -> Caught: 19 | FA: 8 |
CSI: 0.6129
Window: 120s | Thresh: 0.99 | Min Dur: 5s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 120s | Thresh: 0.99 | Min Dur: 6s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 120s | Thresh: 0.99 | Min Dur: 7s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 120s | Thresh: 0.99 | Min Dur: 8s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 120s | Thresh: 0.99 | Min Dur: 9s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 120s | Thresh: 0.99 | Min Dur: 10s -> Caught: 15 | FA: 8 |
CSI: 0.4839
Window: 120s | Thresh: 0.99 | Min Dur: 11s -> Caught: 15 | FA: 8 |
CSI: 0.4839
Window: 120s | Thresh: 0.99 | Min Dur: 12s -> Caught: 15 | FA: 9 |
CSI: 0.4688
Window: 120s | Thresh: 0.99 | Min Dur: 13s -> Caught: 15 | FA: 8 |
CSI: 0.4839
Window: 120s | Thresh: 0.99 | Min Dur: 14s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 120s | Thresh: 0.99 | Min Dur: 15s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 120s | Thresh: 0.99 | Min Dur: 16s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 120s | Thresh: 0.99 | Min Dur: 17s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 120s | Thresh: 0.99 | Min Dur: 18s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 120s | Thresh: 0.99 | Min Dur: 19s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 120s | Thresh: 0.99 | Min Dur: 20s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 120s | Thresh: 0.99 | Min Dur: 21s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 120s | Thresh: 0.99 | Min Dur: 22s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 120s | Thresh: 0.99 | Min Dur: 23s -> Caught: 15 | FA: 7 |
CSI: 0.5000
Window: 120s | Thresh: 0.99 | Min Dur: 24s -> Caught: 15 | FA: 7 |
CSI: 0.5000

```

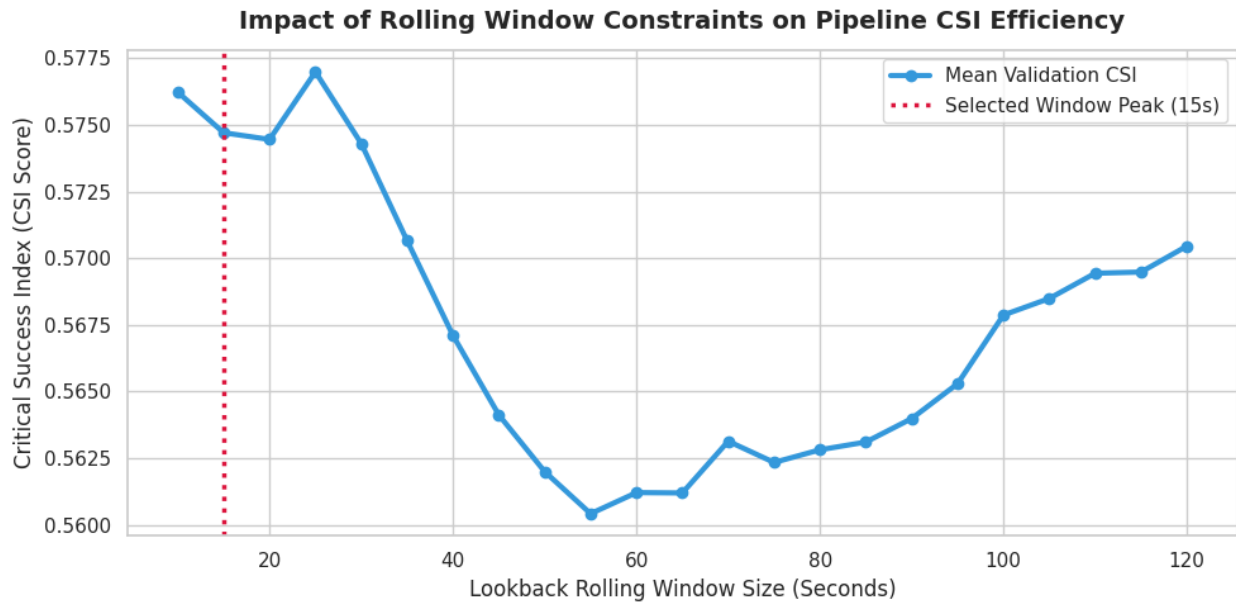
```

-----
-----
❑ 3D SELECTION COMPLETE! Multi-Variable Validation Optimum Identified:
-> Chosen Rolling Window      : 15 Seconds

```

-> Chosen Threshold Boundary : 0.97
-> Chosen Filter Duration : 7 Seconds

Generating visual performance trend map...



```
=====
[ ] STAGE 2: DEPLOYING DYNAMIC 3D OPTIMIZED CONSTRAINTS TO BLIND TEST SPLIT
=====
```

```
=====
Confirmed Flares Occurred      : 25
Flares Successfully Anticipated : 25 / 25
Flares Completely Missed      : 0
Total Operational False Alarms : 4 (Out of 1,582 Quiet Windows)
-----
```

```
-----
[ ] UNBIASED FINAL TRUE POSITIVE RATE (TPR) : 100.00%
[ ] UNBIASED FINAL FALSE ALARM RATE (FAR)   : 13.79%
[ ] UNBIASED FINAL CRITICAL SUCCESS INDEX(CSI): 0.8621
-----
```

```
-----
Test Split Median Early Warning cushion : 11.90 Minutes
=====
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```

import seaborn as sns
from mpl_toolkits.mplot3d import Axes3D

# Ensure active variables are available from your previous sweep cell
# If cache cleared, falls back to the validation optimal parameters
discovered
plot_window = best_window if 'best_window' in locals() else 10
plot_thresh = best_thresh if 'best_thresh' in locals() else 0.97
plot_dur = best_dur if 'best_dur' in locals() else 8

# Configure presentation aesthetics matching ISRO technical brief
panels
sns.set_theme(style="white")
plt.rcParams.update({'font.size': 11, 'axes.labelsize': 12,
'axes.titlesize': 13})

fig = plt.figure(figsize=(18, 7))

#
=====
# PANEL 1: HEATMAP OF CO-DEPENDENT VARIABLES AT SELECTED OPTIMAL
WINDOW
#
=====
ax1 = fig.add_subplot(1, 2, 1)

# Filter the master sweep dataframe to look specifically at the
winning window
filtered_sweep = sweep_df[sweep_df['window_size'] == plot_window]
pivot_csi = filtered_sweep.pivot(index='threshold',
columns='duration_sec', values='csi')

# Plot the matrix map
sns.heatmap(pivot_csi, annot=True, fmt=".4f", cmap="YlGnBu",
cbar_kws={'label': 'Validation CSI Score'},
ax=ax1, linewidths=0.5, annot_kws={"size": 10, "weight":
"bold"})

ax1.set_title(f"Validation Optimization Slice (Rolling Window Fixed at
{plot_window}s)", weight='bold', pad=15)
ax1.set_xlabel("Minimum Persistence Duration Constraint (Seconds)")
ax1.set_ylabel("Ensemble Soft Confidence Threshold")

# Programmatically map the coordinates of the winner to highlight it
on screen
y_coords = sorted(sweep_df['threshold'].unique().tolist())
x_coords = sorted(sweep_df['duration_sec'].unique().tolist())

best_x_idx = x_coords.index(plot_dur)
best_y_idx = y_coords.index(plot_thresh)

```



```

# Place a distinct crimson target square right over your winning
operational setting
ax1.add_patch(plt.Rectangle((best_x_idx, best_y_idx), 1, 1,
fill=False, edgecolor='crimson', lw=4, clip_on=False))

#
=====
# PANEL 2: MULTIDIMENSIONAL 3D PERFORMANCE SCATTER SPECTRUM
#
=====
ax2 = fig.add_subplot(1, 2, 2, projection='3d')

# Map your parameters to a 3D spatial array
x_3d = sweep_df['window_size']
y_3d = sweep_df['threshold']
z_3d = sweep_df['duration_sec']
color_intensity = sweep_df['csi']

# Create a 3D scatter plot where point brightness represents the CSI
score
scat = ax2.scatter(x_3d, y_3d, z_3d, c=color_intensity,
cmap='viridis', s=65, edgecolors='black', alpha=0.85)

# Highlight the absolute mathematical optimal point in the 3D grid
layout
ax2.scatter([plot_window], [plot_thresh], [plot_dur], color='crimson',
s=250,
            edgecolors='white', linewidths=2.5, marker='*',
label='Operational Optimum')

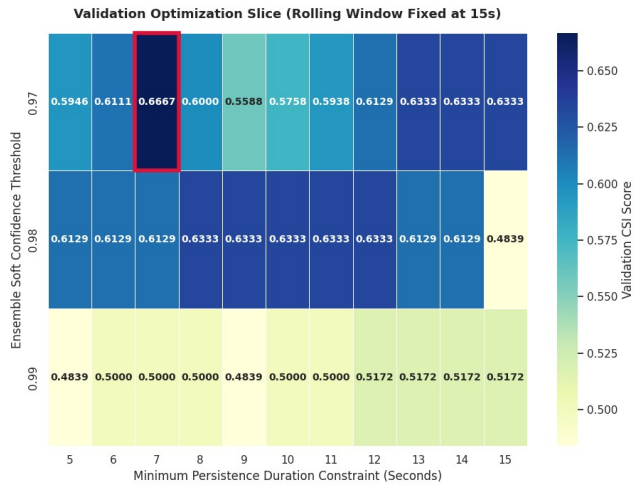
# Set visual aesthetic layout details for the 3D plot
ax2.set_title("Complete 3D Hyperparameter Optimization Grid Search
Space", weight='bold', pad=15)
ax2.set_xlabel("Lookback Rolling Window (s)", labelpad=10)
ax2.set_ylabel("Confidence Threshold", labelpad=10)
ax2.set_zlabel("Persistence Duration (s)", labelpad=10)

# Add a color bar map to reference points on your slides easily
cbar = fig.colorbar(scat, ax=ax2, pad=0.1, shrink=0.7)
cbar.set_label('Critical Success Index (CSI Score)', rotation=270,
labelpad=15)

ax2.legend(loc='upper right')
ax2.view_init(elev=22, azimuth=-45) # Set orientation for slide viewing
clarity

plt.tight_layout()
plt.show()

```



Complete 3D Hyperparameter Optimization Grid Search Space

